

Towards win-win: Evidence from commercial dispute resolution in India

Manaswini Rao, Umer Sohail, Sandhya Seetharaman, and Raghav Pandey *

September 14, 2026

Abstract

Ineffective courts can delay economic development when legal disputes remain unresolved. This is particularly concerning for India where over 50 million suits are pending in courts for more than a year - a statistic that hasn't changed in over a decade. Using random case assignment to judges and policies on mediation, we provide novel causal evidence on the economic benefits of mediation as a dispute resolution strategy for contract disputes in India. Assignment to a "settlement-prone" judge increases dispute resolution by 18 percentage points. Consequently, plaintiff firms' profits increase by 18%, a persistent effect. Conversely, defendant firms' profits decrease by 30% around the settlement timing, although this effect attenuates over time. Reduced uncertainty from dispute resolution, particularly for firms with fewer cases, is a plausible mechanism driving these results.

* Rao and Sohail: U Delaware, Department of Economics (correspondence: manaswini.rao@gmail.com or manasrao@udel.edu). Seetharaman: J-PAL South Asia at IFMR (India). Pandey: National Law University, Delhi (India). Research was conducted with approval from the Institutional Review Boards at the Institute for Financial Management and Research in Chennai, India (IRB # 00007107). We are grateful to Alfred Lerner College of Business and Economics, University of Delaware for Lerner Grant Fall 2024, J-PAL Governance Initiative Multidisciplinary Grant Fall 2023, and Karthik Muralidharan for CEGIS grant Fall 2022 and supporting this project. We are grateful to Khushbu Shah and Mehar Saxena for excellent research assistance. We thank Daksh India team (Surya Prakash B.S., Leah Verghese, and Sandhya P.R.) for engaging discussions on commercial litigation landscape in India. The authors have no financial interests to disclose. This paper was previously titled "Contracts, Dispute Resolution, and Firm Productivity: Evidence from Commercial Courts in India" in funding applications.

1 Introduction

In dynamic business environments, disputes naturally arise due to disagreements on contractual terms, payment delays or defaults, and other disagreements (Banerjee and Duflo 2000; Djankov et al. 2008; Ryan 2020). Economic growth requires effective dispute resolution, where disputes are resolved within a reasonable time and to the satisfaction of the disputing parties (Djankov et al. 2003; Chemin 2020; Chemin et al. 2024). While much of the literature has focused on capacity constraints and downstream economic consequences of judicial delays (Chemin 2010; Ponticelli and Alencar 2016; Coviello et al. 2018; Amirapu 2021; Kondylis and Stein 2023; Rao 2024), we know little about the economic consequences of the dispute resolution process itself. This is important because different economic agents engaged in dispute resolution (the litigants, their lawyers, and the judge) may have often contrasting incentives from judicial delays (Sadka et al. 2024).

This paper examines mediation as a dispute resolution strategy in the context of recently introduced commercial courts in India, random assignment of cases to judges, and policies on mediation during a lawsuit. Using the full universe of legal case data from two large, exclusive commercial courts, we find that mediation-based resolution from being assigned a “settlement-prone” judge improves the effectiveness of dispute resolution. A settlement precludes the often costly, lengthy, and uncertain trial. Subsequently, we find large effects on litigants’ outcomes, where plaintiff firms experience persistent improvements in their profits and production outcomes. In contrast, defendant firms initially experience a negative effect but recover their losses over time.¹

We measure judge-specific settlement propensity as a jack-knife leave-out average settlement rate using cases from other litigants assigned to the same judge (similar to the leniency measure as in Kling 2006; Arnold et al. 2018). We combine the jack-knife instrument with Difference in Differences (DiD) design around the timing of case settlement using a matched balanced panel of litigating firms’ quarterly balance sheet data, separately for plaintiff and defendant firms. We compare changes in the economic outcomes of litigants assigned to a judge who is one standard deviation higher in their settlement-index (“treated”) with those assigned to a judge with an average settlement-index (“control”) before and after the timing of filing and decision of their first settled case in the sample courts. Random assignment of cases and reduced form analysis with DiD design address many of the challenges in causal identification using judge IV designs (Frandsen et al. 2023).²

We verify the random allocation process extensively through documentary checks,

¹Random assignment of cases has recently been built into the Court Information System or CIS as part of the ongoing efforts to digitize and improve the business process workflow in some courts across India. This automatic random assignment in commercial courts started in early 2020.

²Many firms have more than one case in the courts, so we examine outcomes around the first settled case and verify robustness across different cases. Firms with no case that is settled through mediation or those with cases pending for resolution at the end of the study period serve as the never-treated group.

field visits and discussions with court staff, verification based on judge workload from case allocation timestamps, and balance checks across various case-level and litigant-level characteristics. In addition to random assignment, the panel nature of litigants' outcome data from their quarterly balance sheets additionally allows us to test and rule out endogeneity in the timing of case filing and resolution, and rule out differential pre-trends to estimate causal dynamic treatment effects of mediation-based dispute resolution.

Commercial courts were created under the federal Commercial Courts Act, 2015, with the objective of bringing faster resolution to commercial and contractual dispute litigation in India. Following the new law, dedicated commercial courts were established in four commercial centers - Delhi, Mumbai, Bengaluru, and Kolkata - between 2017 and 2020. These metropolitan areas are also regions where over 50% of all formal sector firms in India are located. This study uses data from Delhi and Bengaluru commercial courts following the adoption of random assignment policy in 2020.

Alternate Dispute Resolution (ADR) mechanisms, such as mediation, have been actively promoted world-wide since the mid-20th century (Shavell 1982; Silbey and Merry 1986) and have gained considerable policy traction in countries like India as solutions to severely backlogged courts. Despite the ubiquitous nature of mediation worldwide, there is limited causal evidence on the downstream welfare implications of mediation as a dispute resolution strategy offered by formal courts themselves.³ Recent literature examining informal dispute resolution systems, including the role of local communities (Blattman et al. 2014; Sandefur and Siddiqi 2015; Hartman et al. 2021), documents ambiguous welfare implications with the formal system often acting as a binding constraint (Mattsson and Mobarak 2025).

This paper addresses this gap by leveraging random assignment of cases to judges in formal courts of law and policies on mediation such as Section 89 of the Code of Civil Procedure where the judge assigned to the case may continue to recommend mediation-based settlement at any stage during the lawsuit. The typical process of a commercial lawsuit is as follows: a plaintiff files a case, the case is randomly assigned to a judge by a computerized system, the judge summons plaintiff and defendants for their first “case management” hearing where the judge may discuss mediation as a potential dispute resolution strategy. If mediation is successful, the case is resolved and exits the pending backlog. If mediation fails, full trial process begins.

We document several results. First, we find a strong first stage. The probability that a case itself is settled through mediation is largely determined by the judge assigned to the case. Judges vary in their settlement propensity; some judges are better at communicating about mediation and uncertainties in the trial process whereas others are not. In typical

³Theoretical work, for example Shavell 1982, 2026 shows that the welfare effect of mediation can be mixed due to differential litigation strategies adopted by plaintiffs and defendants in response to settlement as an option. Another vast literature in socio-legal studies and sociology of law documents mixed effects of mediation using qualitative methods such as Silbey 1998; Wissler 2004.

case management hearings, judges discuss the details of the case with the litigants and provide information on the legal processes. Judges vary in what type of information they provide and how they provide (Silbey and Merry 1986), which could affect the litigants' decision to settle or not. We find that a case assigned to a judge with one standard deviation higher settlement index is close to 18 percentage points more likely to be settled. This probability is similar whether we examine cases where the sample firm appears as a plaintiff or when they appear as a defendant.

Second, we find systematic correlation between resolution of a case by settlement and the efficiency of the judicial process measured in terms of case duration or pending status. A case that is settled is associated with 50 fewer days pending in court on average. The difference in the distribution of case duration is even more stark between cases that are settled through mediation and those that continue with trial. This also has implications for pending backlog - cases assigned to judges with one-standard deviation higher settlement index are 8.4 percentage points less likely to remain pending by the end of the study period.

Third, we find significant heterogeneity in settlement responses by financial sector firms. Specifically, financial sector plaintiffs are 19 percentage points more likely than non-financial plaintiffs to settle a case during mediation on their own. The additional effect of being assigned a settlement-prone judge among financial sector plaintiffs is relatively muted. These differential firm-level settlement behavior are associated with additional judicial efficiency effects such as lower case duration for financial sector plaintiffs.

Fourth, we find real economic effects on litigating firms. We find 18% increase in quarterly profit among plaintiff firms assigned a judge with one-standard deviation higher settlement index. Among defendants, we find about 30% decrease in quarterly profits. These effects are seen immediately following case filing when mediation is offered rather than around the timing of formal case decision (date of the verdict). The profit effects persist for plaintiff whereas the negative effect partially reverses for defendants. Much of the positive effect on profits among plaintiff firms is driven by those with lighter caseloads, for whom the dispute could plausibly represent a holdup. Consequently, we find that treated plaintiff firms expand their wage bill by 15% and long-run investment by 5% after the formal case decision date. We find negative but noisy effects among defendant firms.

The timing of the profit effects and the longer run expansion decisions by plaintiffs suggest that the results could plausibly be driven by reduced uncertainty from avoided trial (Bamieh et al. 2025). We also observe increased recoveries among financial sector firms assigned to settlement-prone judges. These results are consistent with a large literature in corporate finance showing that current period liquidity has a higher marginal value in the context of missing capital and credit markets (Faulkender and Wang 2006; Djankov et al. 2008; Cai and Szeidl 2024).

We frame our findings as a ‘towards win-win’ outcome that operates on two levels. First, settlement leads to efficiency gains to the process of dispute resolution by reducing case duration and pendency. Second, settlement offers a weakly Pareto-improving outcome for litigants: plaintiffs experience substantial gains, while the initial loss experienced by defendants are partially recovered post case decision. The counterfactual is that these litigants would otherwise undergo lengthy and uncertain trial process in the absence of mediation, where the trial outcome could even be worse than settlement. We situate this framing within the specific context of commercial litigation landscape where firms engaged in such litigation are large and approach courts as the last resort. We do not find any evidence that plaintiff firms that settle their case through mediation are more likely to file additional new cases in courts. In contrast, the evidence suggests that initial experience of settling is correlated with increased settlement in subsequent cases.

We carry out multiple robustness tests to verify these results. First, we find that the profit effects are robust even under different restrictions to the event window. Second, we account for the fact that a firm can have multiple cases by focusing on the first settled case. We verify that the effects are robust using later settled cases. Lastly, to the extent feasible, we rule out the possibility that these effects are not driven by other judge characteristics such as judge’s human capital measured in terms of their ability to cite other case precedents, the value of the case (“claim”) assigned to the judge, or the recovery rate (award amount as a percentage of claim).

This paper contributes to the literature on dispute resolution by courts and its development ramifications by documenting *what* improves dispute resolution. This is among the first set of papers in this literature to use judge-specific characteristics, such as settlement propensity, to resolve a contractual dispute in a timely manner as opposed to procedural or legal reforms that are often complicated and require support through legislation. The results suggest improvements in the efficiency of dispute resolution as well as in the profitability of firms engaged in commercial disputes. This is consistent with a rich theoretical and empirical literature on court efficiency and economic growth, suggesting that firms and economic agents shift to a second-best equilibrium in response to judicial inefficiency (Djankov et al. 2003; Nunn 2007; Visaria 2009; Coviello et al. 2015; Ponticelli and Alencar 2016; Coviello et al. 2018; Chemin 2020; Amirapu 2021; Liu et al. 2022; Kondylis and Stein 2023; Bamieh et al. 2025). This paper provides experimental evidence from random assignment of cases to settlement-prone judges, showing the inverse of this relationship: when courts function better, firms benefit. This paper highlights the importance of mutually agreed-upon settlement of the dispute, facilitated within the legal framework by a judge, as a solution to the problem of court inefficiency.

Furthermore, we show that financial institutions such as banks respond strategically to mediation. We show that banks are among the biggest users of courts for contractual disputes concerning their lending operations and prefer a faster resolution of disputes as

plaintiff. In contrast, they are less likely to settle a case as a defendant particularly in cases that are counter-suits filed by borrowers to prevent banks from liquidating assets. Consequently, banks are potentially less likely to agree with settlement conditions in such situations. Thus efficiency of improving the dispute resolution process specifically matters for financial sector plaintiffs in debt recovery (Visaria 2009; Lilienfeld-Toal et al. 2012; Rao 2024). Settlement could be an effective debt contract enforcement strategy when trials take longer relative to the implicit discount rate on the claim amount in debt recovery litigation (Djankov et al. 2008).

The results in this paper are also consistent with Silveira 2017; Sadka et al. 2024 on the role of litigant over-optimism in their trial outcome, and with Bamieh et al. (2025) on the role of uncertainty from trial duration. The mediation process addresses some of the key challenges arising from information asymmetry and over-optimism, enabling firms to resolve their dispute through mutually agreeable settlement terms, facilitated by judges, rather than continuing full trial. Avoiding trial could also reduce uncertainty from longer trial duration. This paper thus extends the scope and conclusion by Sadka et al. (2024) and Bamieh et al. (2025), who study labor disputes, and Silveira (2017) examining criminal plea bargaining, to commercial disputes in the context of a developing economy. Settlement generates immediate liquidity for plaintiff firms. This additional cash at hand has a higher marginal value in developing countries with limited access to capital markets (Faulkender and Wang 2006; Cai and Szeidl 2024).

The rest of the paper are organized as follows: section 2 provides background about the policy, section 4 details the research design to estimate the causal effects of being assigned a settlement-prone judge on case and firm-level outcomes. We details the datasets and analysis sample construction in section 3, discuss the results in section 5. We discuss the results in light of doctrinal legal analysis to suggest policy action, particularly to the legal and judicial stakeholders in the policy space in section 6, finally concluding in section 7.

2 Background

Prior to 2015, commercial litigation in India was handled by general civil courts, which were often overburdened with cases across different dispute types. This resulted in significant delays in adjudicating commercial matters, undermining contractual reliability and increasing transaction costs for firms operating in the country (Ghosh, 2018).

In response to these inefficiencies, the Commercial Courts Act, 2015 was enacted by the Government of India to establish dedicated courts for the adjudication of commercial disputes. The Act introduced Commercial Courts at the district level, Commercial Divisions in High Courts with original jurisdiction over very large value cases, and Commercial Appellate Divisions for appeals. According to Section 2(1)(c) of the Act, the scope of a “commercial dispute” includes a broad set of issues such as breach of contract,

shareholder disputes, intellectual property rights, and admiralty matters, provided the monetary value involved exceeds *Rs.*1 crore (approximately USD 120,000).

The legislation aimed to streamline dispute resolution through procedural innovations. A key feature is mandatory mediation, which must be undertaken before the dispute is officially filed as a commercial lawsuit — except in cases requiring urgent interim relief. Another relevant innovation was the introduction of random allocation of cases to judges through a digital Case Information System (CIS 3.2). Since 2020, this system has been operational in the dedicated Commercial Courts of Delhi, Mumbai, Bengaluru, and Kolkata. Automatic assignment of cases without human intervention was intended to minimize any scope for “forum shopping” or bias in judicial assignment, with a stated objective to enhance transparency and fairness in the process.

All of these improved processes and reform continue to function within the broader procedural legal framework of the Indian law governed by the Code of Civil Procedure, 1908 (CPC). Under this, litigants do not have a choice over which court to file their lawsuits in. This process is governed by specific sections of CPC that specify territorial and monetary jurisdiction of courts.

Once a dispute is filed as a suit, the process is as follows: a plaintiff files their dispute as a lawsuit in the concerned commercial court. The court registrar uses the in-built randomization tool to assign the case to a judge. Once the case is assigned, the judge summons both the plaintiff and defendants for a case management hearing. The judge discusses the main aspects of the dispute, provides information on mediation as alternate dispute resolution strategy (which is also governed by Section 89 of CPC), and also provides information on what the litigation process could look like if mediation fails. In the event of mediation failure, the formal trial process begins with structured phases: filing of pleadings, disclosure and inspection of documents, issue framing, examination of witnesses, and final arguments.

On paper, the judgment for suits undergoing full trial is required to be delivered within 90 days after the conclusion of final arguments. However, in practice, the examination phase remains the most time-consuming, often extending up to six months, making the entire process take more than a year (The Economic Times, 2021). In contrast, mediation has the potential to resolve the dispute within a single or at most a few hearings.

This paper focuses on the effects of mediation once a commercial dispute is filed as a lawsuit in two of the four commercial courts with random assignment of cases.

3 Data

This paper combines two primary data sources: (i) firm-level quarterly balance sheet data from the Prowess dataset curated by the Centre for Monitoring Indian Economy (CMIE), and (ii) universe of case-level records from Delhi and Bengaluru commercial

courts, following the adoption of random assignment policy in 2020 spanning 3 years (2020-2023). This section describes each dataset, outlines how we construct the analysis dataset (firm–judge panel), and details our various classification algorithms to generate variables of interest.

3.1 Firm-Level Data

We use firm-level panel data from Prowess, spanning formal sector Indian firms. The dataset covers publicly listed, unlisted, and privately held companies and includes annual and quarterly information on profit and loss statements, balance sheets, cash flows, and firm characteristics such as industry affiliation, ownership, and incorporation year. Our primary firm-level outcome variables are quarterly profit (income net of expenses), wagebill, and investments.

3.2 Judicial Data and Entity-Type Classification

We compile a case-level dataset of the universe of 6624 commercial court filings in Delhi and Bengaluru commercial courts since the start of random assignment of cases. These data contain detailed records for each case, including plaintiff and defendant names, filing and resolution dates, and basic metadata such as the judge name and case type.⁴

To structure the data for empirical analysis, we develop a rule-based text classification algorithm to categorize both plaintiffs and defendants into one of three mutually exclusive entity types: Individuals, Firms, and Banks.

Firms are identified using a large set of regular expressions and keywords commonly associated with companies and organizations (e.g., LTD, LLP, PVT, TECH, FOUNDATION, SOCIETY, TRUST, etc.). Banks and financial firms are further segregated from general firms using keywords such as BANK, FINANCE, CREDIT. All remaining entities are classified as individuals, assumed to be natural persons not associated with any institutional identifier.

This classification is applied separately to plaintiffs and defendants. We then construct a categorical variable for each case indicating the entity-type pair involved in the dispute (e.g., Firm–Firm, Bank–Individual). This approach enables scalable classification of cases by type.

3.3 Matched Panel Construction

We link commercial court cases to firms in Prowess by matching on the names of plaintiffs and defendants. We use a string-matching procedure to compare the cleaned names in

⁴We also have the universe of all case-level data from the inception of these courts but we do not include cases that were not randomly assigned to judges in our analysis.

court data with firm names in Prowess. The match is based on exact or near-exact string matches, allowing for minor spelling or formatting differences. We keep only those cases where the match quality is high—specifically, where the string similarity score exceeds 90%. This ensures that we are linking firms to court cases with a high degree of confidence. We manually verify the resulting match to ascertain its correctness.

Using this approach, we construct two matched samples: one where the firm appears as the plaintiff and one where the firm appears as the defendant in a court case.

3.4 Summary of Entity-Type Pairs

Panel A of Table 1 presents the distribution of entity-type pairs across the full case-level sample and matched plaintiff and defendant samples. In the full sample, firm to firm cases represent a third of all commercial disputes (33.45%). Banks as plaintiff cases represent another third of the disputes, followed by firm–individual (14%) and individual–firm (10.52%).

The matched plaintiff sample is overrepresented by bank-related cases (bank-individual (33.8%) and bank-firm cases (46.95%), while firm-firm (12.86%) or firm-individual (5.96%) cases are under-represented. The defendant-matched panel contains a higher share of firm to firm disputes.

Table 1 Panel B presents summary statistics for firms in the full Prowess dataset and for firms matched as plaintiffs or defendants in our sample. Firms in the matched plaintiff sample are, on average, slightly older (37.5 years) than both the general Prowess population (30.2 years) and firms in the defendant sample (32.8 years). This suggests that firms initiating litigation tend to be more established.

Sectoral composition is representative for trade and services sectors whereas manufacturing is underrepresented. The analysis sample also over-represents the set of publicly listed firms relative to their share in the Prowess sample.

These patterns plausibly occur due to differential composition of Prowess database relative to firms involved in lawsuits. Since Prowess is among the available datasets with high frequency firm-level outcome data, we caveat all inference keeping this limitation in mind.

3.5 Case Type Composition

There are four types of cases in commercial courts: (a) Execution (ComEX), (b) Appeals (ComAA), (c) Original/New Suits (ComOS), and (d) a residual Other category. Table A1 reports the average shares of these case types across the full case sample and the two matched samples. In the full sample, nearly half (48%) of all cases are original suits, followed by execution proceedings (23%) and other case types (19%). Appeals constitute

a smaller share (10%). The analysis sample for plaintiff firms maintains this ordering qualitatively, where original suits make the largest share, followed by execution.

3.6 Analysis Sample

We carry out our analysis separately for firms appearing as plaintiff and firms appearing as defendants. The plaintiff sample has 1490 cases matched to 154 unique firms in Prowess with quarterly balance sheet data. The defendant sample has 248 cases matched to 124 unique firms. Within the plaintiff sample, over 85% plaintiff cases map to 35 unique financial sector firms and the remaining map to 119 unique non-financial firms. In the defendant sample, 53% cases map to 41 unique financial sector firms and the remaining 47% cases map to 83 non-financial sector firms. The firm-level quarterly data span 2011-2024 for each of these sets of firms.

4 Empirical Design

We leverage random assignment of contract and commercial dispute cases to judges in Bengaluru and Delhi commercial courts. This new randomized case assignment system (see Figure A1 for an example of the assignment process) that started in 2020 replaced an old system, which was a black-box that followed the discretion of court’s principal administrative judge in assigning cases. The main case-level data used in the analysis for this paper includes all new cases that were randomly allocated to the judges in these sample courts as described in section 3.

Following a large literature in empirical legal studies and labor economics such as (Kling, 2006; Arnold et al., 2018), our empirical design combines judge-leniency design with difference in differences (DiD) design. The “treatment” is whether a case is assigned to a settlement-prone (“lenient”) judge or not, which we measure as the jack-knife leave-out settlement average using outcomes of cases by other litigants presided by the same judge.

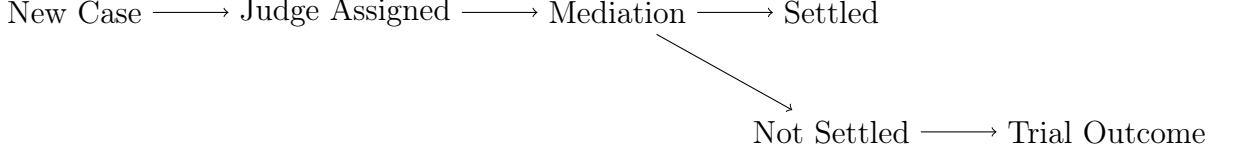
We construct the treatment variable as below, following the standard leave-out judge leniency instrument as constructed in Arnold et al. 2018:

$$z_{cj} = \left(\frac{1}{n_j - n_{ij}} \right) \left(\sum_{k=0}^{n_j} \text{Settled}_{ik} - \sum_{c=0}^{n_{ij}} \text{Settled}_{ic} \right) \quad (1)$$

where Settled_{ik} represents a dummy variable for each case k with a judge j that takes value 1 if that case is settled. Similarly, Settled_{ic} is a dummy variable for firm i ’s case c with judge j that takes value 1 if it is settled. n_j and n_{ij} are the total number of cases assigned to judge j and the number of firm i ’s cases assigned to judge j , respectively. z_{cj} is the leave-out average proportion of cases assigned to judge j that are settled. Due to

data limitation, we compute the leave-out settlement rate at judge level as opposed to judge-year level to preserve statistical power.

The random assignment of newly filed cases to judges and mediation options introduce the following causal chain.



Following this, cases are either: (a) resolved during mediation (“settled” through reconciliation between the two litigating parties), (b) fails mediation and resolved at the end of the trial, and (c) fails mediation and is pending for resolution. We estimate the following empirical specification as the “first stage” of contract enforcement in courts:

$$\text{Case Outcome}_{c(i)j} = \delta_s + \delta_m + \delta_l + \beta z_{c(i)j} + \varepsilon_{c(i)j} \quad (2)$$

where $\text{Case Outcome}_{c(i)j}$ of case c of firm i assigned to judge j includes whether the case is resolved through settlement. It can also mean other outcomes including duration and pending status. The specifications include city location s (or the court) fixed effect, registration month m fixed effect, and case type l fixed effect. City location and registration month are stratifying variables used in the randomization process and thus, follow the design. We also include case type l fixed effect (for example, if some case types are more amenable to mediation) and test for robustness with and without. We standardize the leave-out instrument in standard deviation units so that we infer treatment as being assigned to a judge with one standard deviation unit higher settlement index relative to a judge with average settlement index. We cluster the standard errors by judge, which is the unit of treatment variation.

We examine the outcomes of the litigating firms, separately for plaintiff and defending firm samples. Taking advantage of panel structure of the balance sheet data, we estimate the effects of being assigned a settlement-prone judge on firm productivity. We execute this as a generalized (continuous-valued) DiD design using the leave-out settlement average before and after the quarter the corresponding case is resolved in the court. Our estimating equation is as follows:

$$Y_{it} = \phi_s + \phi_m + \phi_i + \phi_t + \gamma z_{c(i)j} \times \text{Post}_t + \alpha \text{Post}_t + \epsilon_{it} \quad (3)$$

where i denotes the litigating firm with outcome Y_{it} reported in quarter t . Post_t

indicates quarters after the last date (“decision”) of the first settled case.⁵ The rest of the subscripts are as defined as before. In addition to city location and registration month fixed effects, we include firm and reporting quarter fixed effects as in standard DiD specifications. As before, we standardize the judge instrument in standard deviation units. We cluster standard errors by the assigned judge.

Using Equation 3, we compare the outcomes of firms assigned to a judge with one standard deviation higher settlement index to those assigned to judges with average settlement index before and after their corresponding case is resolved. We also include firms whose cases continue to remain pending until the end of the study period as well as cases that are resolved through trial instead of mediation as never-treated.

For causal inference, we invoke additional assumptions used in staggered DiD designs as the dates on which a case is settled are staggered for different litigants. Random assignment of cases ensures that the judge leave-out settlement index is exogenous to case and litigant-specific potential outcomes. Since we estimate our main models - Equation 2 and Equation 3 - as reduced-form specifications using OLS estimators rather than estimating the local average treatment effect (LATE) parameter using two-stage-least-squares (2SLS) estimator, the coefficients should be interpreted as the average treatment effect on the treated (ATT). This also addresses the problem of selection bias arising from selection to different treatment intensities (Callaway et al. 2024) and reduces concerns about negative weights, although estimating Equation 3 still assumes homogeneous treatment effects. We present the raw means and include multiple event indicators to address concerns about dynamic treatment effects.

We use a variety of tests to address concerns about exclusion restriction and monotonicity. For exclusion restriction to hold even with the random assignment of cases to judges, we need to assume that judges affect both plaintiff and defendant outcomes only through settlement of *their specific case* during mediation and not through other channels (such as firms’ overall litigation strategy). While this is a fundamentally untestable assumption, the workflow of a commercial lawsuit includes mediation typically as the first step for cases filed in these courts. The implicit assumption then is that once a litigant encounters a settlement-prone judge, they do not alter their litigation strategy across other contractual disputes. We discuss this in detail later in this section.

Recent literature (Bhuller and Sigstad 2022; Sigstad 2023; Frandsen et al. 2023) has questioned the validity of judge leniency designs satisfying the monotonicity assumption in studies examining the consequences of sentencing or bail decisions, which implicitly impose a single dimension of decision-maker behavior (such as acquittal or bail approval). With the caveat that it is impossible to observe any judges’ internal ranking on case outcomes in the context of commercial disputes, settlement through mediation additionally requires agreement by the litigating parties. Since the litigant type and the dispute type

⁵Since a firm can have multiple cases, we test for robustness using other cases of the same firm.

are, by design, exogenous to judge’s characteristics, it is plausible that if a less settlement-prone judge helps settle a case, then a more settlement-prone judge should also help settle the same case if it was assigned to them.

4.1 Balance Test

Table 2 presents the results of balance tests using settlement index of assigned judge as the treatment variable. Columns 1-3 present omnibus test of exogeneity using judge id as the dependent variable with explanatory variables including types of litigation - whether it is a bank vs. bank, bank vs. firm, firm vs. firm, and so on - and litigating firm characteristics (when we are able to identify and match the firm with firm-level data). Col 1 includes the entire case-level data in the commercial court sample. Columns 2-3 are restricted to the matched firm-level analysis sample of plaintiffs and defendants, respectively.

Columns 4-5 report results from a similar exercise where the dependent variable is the leave-out settlement rate. These columns present the required tests of exogeneity since settlement rate and not judge id is the exogenous variable. While some individual case and firm-level characteristics are statistically significant, we are unable to reject the joint null of any case or litigant-specific characteristic determining judge settlement propensity. Moreover, we account for all firm-specific time invariant potential confounders in the form of firm fixed effects in our empirical specifications.

We also verify random assignment of cases to judges by examining the properties of its statistical distribution. Assuming that the random assignment follows a uniform distribution so as to generate equal workload among the judges, we find that the correlation between assigned number of cases and expected number of cases per judge on any given day is close to 1 (see Figure 1).

4.2 Exclusion Restriction

While randomization ensures exogeneity between a judge’s settlement propensity and potential outcomes, it does not guarantee exclusion restriction. That is, being assigned a settlement-prone judge should affect the litigants’ outcomes only through the fact that the specific case is settled and not because of other induced behavior that is also correlated with the judge’s settlement propensity. For example, instead of settlement in the corresponding case, if the experience of being assigned to a specific judge alters the firms’ overall litigation strategy or contracting behavior, then exclusion restriction is likely violated.

Table A2 examines the correlation between the settlement propensity and other metrics reflecting the “quality” of the judge including number of pages in the judgment orders, and citation of precedents using text of judge orders from the case-level analysis sample.

We don't find evidence of any substantial correlation, suggesting that judge-specific human capital is unlikely to violate exclusion restriction. We also do not find substantive evidence that lower-valued cases are more likely to be assigned to settlement-more judges.

Litigating firms are also no more likely to file new lawsuits following the settlement of their first few cases, suggesting that such firms are unlikely to alter their litigation strategy. See subsection 5.5 and subsection 5.6 for further discussion of exclusion restriction.

5 Results

In this section, we first detail the effects of being assigned a settlement-prone judge on case-level outcomes and subsequently discuss the effects on litigating firms' outcomes.

5.1 Judge Settlement Propensity

We calculate the main explanatory variable - judge settlement index - following the construction process described in Equation 1 separately for plaintiff and defendant samples, respectively. Figure A2 depict judge-level variation across plaintiff and defendant samples. The difference in the distributions of settlement index is plausibly due to differences in the composition of judges in each dataset as well as differences in the composition of firms as a plaintiff or defendant.

5.2 Case-level outcomes

First, we find a strong first stage relationship between the judge leave-out settlement rate and the case-level settlement indicator. We find that being assigned a judge with one standard deviation higher settlement index is strongly correlated with a case being settled in both plaintiff and defendant datasets. Plaintiff firms' cases are 18 percentage points more likely to be settled (Col 1 Table 3) and defendant firms' cases are 22 percentage points more likely to be settled (Col 3 Table 3). Judge settlement index has a substantial explanatory power, with F-statistic > 150 in the plaintiff dataset. This statistic is smaller in the defendant dataset, likely due to a smaller matched sample of cases.

We find significant heterogeneity in case-level outcomes depending on the sector of the litigating firm. Columns 2 and 4 Table 3 present the results from examining heterogeneity by whether a litigating firm is a financial sector firm or not. We find that financial plaintiff firms are 19 percentage points more likely to settle irrespective of the judge assigned, suggesting their settlement-oriented dispute resolution strategy. We find muted interaction effect of judge settlement index among financial sector plaintiff firms. As defendants, financial sector firms are no more or no less likely to settle on their own whereas the interaction effect is negative and statistically significant.

These differences suggest potentially different bargaining power based on the litigant identity (plaintiff vs defendant) of the firms. Financial sector firms are defendants in cases where borrowers file counter-suits to prevent liquidation or restructuring in debt recovery disputes. In such cases, financial firms are less likely to settle as that would mean withdrawing their liquidation or asset restructuring processes. In contrast, such firms are more likely to renegotiate parts of contractual terms, such as waiving parts of interest dues or extend repayment period as plaintiff in debt recovery litigation.

Second, we find strong correlation between case settlement through mediation and the “efficiency” of dispute resolution. Panel A Table 4 presents correlations between case settlement and its duration and pending status, separately for plaintiff and defendant samples. Overall, we find that when a case is settled through mediation, it is around 16 and 23 percentage points less likely to be associated with pending status across plaintiff and defendant samples, respectively. We also note that a settled case is associated with lower duration from the time of filing. On average, plaintiff cases are settled about 50 days earlier than cases that are not settled and go through full trial. Among defendants, this association is even larger: settled cases experience 137 fewer days in court relative to other cases. The distributional effects on case duration is much more stark relative to the average treatment effects. Figure 2 shows the kernel density distributions of case duration in both plaintiff and defendant datasets by their settlement status, highlighting a significant leftward shift among cases that are settled.

The reduced form effects are also in similar direction although lose statistical precision for case duration. Being assigned a judge with one standard deviation higher settlement index results in the case being about 8 percentage points less likely to be pending among plaintiff firms’ (see Panel B Table 4). For defending firms, these numbers are higher.

5.3 Firm-Level Productivity Effects

In our matched data for plaintiff firms, only 10% of the firms have just one case. The median non-financial sector firms like those in manufacturing or trade have 1 case per firm (average number of cases is 1.46; the maximum number of cases per firm in this sector is 5). In contrast, the median financial sector firm such as a bank has 72 cases per firm, with a maximum of 471 cases. Among the defendant sample, 25% of the firms have only one case (median is 2 cases per firm). The median non-financial sector defendant firm has 1 case (average is 1.55 and maximum is 7 per firm). The median financial sector defendant firms have 3 cases, with maximum 25 cases per firm.

Since firms can have multiple cases, the main analysis using firm sample uses the timing of the first settled case, with robustness tests using the timing of other settled cases.

We define a firm as “treated” (i.e., resolve their dispute through settlement) if the

earliest settled case (in terms of date of decision) is assigned to a judge with a one standard deviation higher settlement index. The timing of treatment is the filing and/or decision dates of that case. That is, we define two post-treatment periods - first are all quarters following the date of filing of the case and the second is all the quarters following the decision date of the earliest settled dispute in the courts. For firms with only one case, this corresponds to the outcome of that case if that case was settled through mediation. This naturally generates a staggered treatment design with pre- and post periods in quarterly profits relative to the date of case decision. The never-treated group includes all firms assigned judges with average settlement index as well as those with their case decision still pending. We test for robustness using alternate definitions of treatment, including using timing of the last filed case that is settled.

Figure A3 presents the distribution of quarterly profits among the sample firms, separately for plaintiff and defendant firms, prior to the random case assignment policy in 2020. The average baseline quarterly profits of plaintiff firms is *Rs.* 504 Million (approx USD 5.7 Million) and that of defendant firms is *Rs.* -220 Million (USD -2.5 Million). The differences in baseline profits between groups of firms with settled cases vs those without are not statistically significant.

Figure 3 depicts the raw trends in the levels of firms' quarterly profits before and after the filing date and judges' decision date by groups based on whether their case is settled through mediation or not. The left panel uses the full event window around the timing of filing the lawsuit. The right panel narrows the event window to start from the filing date of the lawsuit and continuing 3 years post the decision date. We note that the profits of plaintiff firms are higher after filing⁶ for the group with settled cases relative to group with cases that are either not settled or continue to be pending at the end of the study period. In contrast, we note the opposite result among defending firms. Since the mediation outcome of a specific firm is potentially endogenous, we examine the reduced form effects of being assigned a settlement-prone judge, measured using the jack-knife leave-out measure. Following reduced form analysis, the main policy variable is the "leniency" of the judge assigned to a specific case and not the outcome of the case as settled or failed to settle. Therefore, we interpret all the subsequent results as what happens to the outcomes of firms whose cases are assigned to a settlement-prone judge.

We report the DiD estimates among plaintiff and defendant firms, respectively, in Table 6. Odd-numbered Columns present the results from the plaintiff dataset whereas even-numbered Columns present the results from defendant dataset. We include time dummies for the date of filing as well as the date of decision of the case in order to understand the dynamics of the treatment effects. The quarterly profit of plaintiff firms increases substantially by 18% after being assigned to a one standard deviation higher settlement-prone judge, which is immediately evident after filing and continues to remain

⁶Mediation is typically offered soon after filing during the first case management meeting

high after resolution relative to firms assigned to judges with average settlement index. For defending firms, the effect of a settlement-prone judge on quarterly profits is negative after filing, but this effect wears away subsequently.

Table A6 presents robustness against the definition of settlement when firms have multiple cases. We find qualitatively similar effects even when we examine the consequences of settlement among the last settled case.

5.4 Reduced Uncertainty from Lawsuits and Production

Next, we focus on wage bills and investments to examine subsequent production decisions. When a firm file a lawsuit, they may reduce labor demand or investment decisions if there is significant uncertainty in the outcome of the lawsuit. The reason for the lawsuit could also be hold-up: a firm files a lawsuit in response to a broken contract. When such disputes are settled through mediation, broken contracts are renegotiated, allowing firms to undertake production decisions in the face of reduced uncertainty. This could result in an increase in input demand and increased long-term investment in productive capacity.

Plaintiff firms assigned to a settlement-prone judge expand both their wage expenditures and investment following the decision on their case relative to those assigned judges with average settlement rate. Conversely, defendant firms assigned to settlement-prone judges contract their wage expenditures and investments following their case decision although the estimates are noisy and lack statistical significance (see Table 7 and Table 8).

In Table 9, we present the details of case-specific judgment orders using text analysis. We find that settlement terms favor plaintiff firms in situations where defendants are absent (“ex-parte” decision). The value of the decree issued by a settlement-prone judge is substantive - over 80% in value relative to cases assigned to judges with average settlement rate.

The effect of reduced uncertainty is particularly evident for financial sector firms who are able to recover bad debt following settlement in commercial courts.⁷ In Table 10, we present the heterogeneous effects of being assigned a judge with one standard deviation higher settlement index among financial sector firms on the extent of recoveries (a reduction in write-offs) by the timestamps of the first settled case. The effects are relatively smaller and insignificant for non-financial firms but substantially significant (in levels as well as in log) for financial sector firms in quarters following mediation-based settlement of their case assigned to a judge with one standard deviation higher settlement index.

⁷Although there is no direct measure called recoveries in Prowess data for financial firms, we use “write back of provisions” as a proxy for recoveries.

5.5 Firms’ Subsequent Litigation Behavior

An unintended consequence of increased resolution of lawsuits through mediation could mean an increase in future lawsuits. If this were to be the case, it may crowd-out business as usual resolution through other means and increase the demand for court-based dispute resolution and even reduce overall judicial efficiency. However, we do not find strong evidence supporting this explanation.

First, we do not find that firms “crowd-in” new litigation following settlement of their lawsuits through mediation. The number of newly filed lawsuits is relatively flat over time among plaintiff firms (see Figure A5). Second, we observe that such firms are more likely to settle their subsequent lawsuits through mediation. This suggests that mediation is potentially an efficiency-improving service provided by the judiciary, which is unlikely to crowd-out business as usual dispute resolution outside the judicial system while at the same time holds potential to reduce pending case backlog and reduce uncertainty involved in litigation.

5.6 Exclusion Restriction

In order to interpret the effects as due to settlement propensity and not due to any other characteristics of the judge that could be correlated with their settlement propensity, we show that settlement-prone judges are not any more likely to rule in favor of plaintiffs during a trial (when mediation fails). Table A4 documents the effect of being assigned to a judge with one standard deviation higher settlement index on plaintiff-win as the outcome. A plaintiff-win is measured using the detailed judgment order text file and is defined as a dummy variable with value 1 when the suit is “allowed” or “decreed” by the judge with decree favoring the plaintiff. Other case outcomes include case “closed” or “dismissed”/“disposed”/“rejected” for lack of a substantive issue, which indicate a decision that does not favor the plaintiff (plaintiff-win is coded as a 0). Cases can also be withdrawn by the plaintiff, in which case the plaintiff-win is coded as a 0.⁸

We find that cases assigned to settlement prone judges are not any more likely to favor plaintiff during trials. In fact, most of the plaintiff wins occur during settlement during mediation, which results in an enforceable decree. We similarly find that cases assigned to such judges are not anymore likely to favor financial sector plaintiff in their trial outcome.

In Table A5, we discuss the details of who goes through the trial process and whether this is correlated with the judge settlement index. We don’t find any support for this type of strategic behavior by the litigants to be correlated with judges’ settlement index.

⁸We used Claude Opus 4.6 model to read all text information from the final orders scraped from court websites as PDF documents.

5.7 Treatment Effect Size

The relatively large profit effect among plaintiff is mainly driven by plaintiffs with fewer cases (Figure 4). For such firms with fewer cases, resolution of their case through mediation potentially unlocks a meaningful constraint relative to those firms with multiple cases (for example, large national banks). Plaintiff firms with 2 or fewer cases in commercial courts are more likely to be from trade, non-financial services, and manufacturing sectors relative to banks or financial services (Figure A6). For such firms, settlement of disputes through mediation is associated with large positive profit effects. For plaintiff firms with more cases, such as banks, the average marginal effect is muted and noisy. However, even such firms benefit through other means such as increased recoveries in Table 10. The negative effects among defendant firms are almost uniformly negative by their underlying number of cases per firm.

The wage effects are almost homogeneous across firms within the plaintiff and defendant samples, respectively (Figure A7). All plaintiff firms experience an increase in their wage bill following settlement through mediation, irrespective of the number of cases in the court. Similarly, all defendant firms experience a contraction in their wage bill following settlement.⁹

5.8 Interpreting LATE

Given our research design exploiting random assignment of cases to judges with different settlement propensities, the interpretation of the causal effects estimates merits a discussion on the differences in the characteristics between compliers and the analysis samples. Table A3 presents the complier and sample means for various characteristics for the plaintiff and defendant samples, respectively. In the plaintiff sample (Panel A), firms with fewer cases, those belonging to the non-financial sector, and older firms are slightly overrepresented among compliers. The complier means are relatively similar to the sample means for other characteristics, suggesting that the complier firms are similar to the average firm in the matched plaintiff sample.

In contrast, financial sector firms are a lot underrepresented among compliers in the defendant sample, consistent with a different strategy towards settlement as defendants (Panel B). Older, listed and non-financial firms are overrepresented as compliers in the defendant sample.

⁹Though the number of cases could potentially be endogenous, I show in Figure A5 that judge settlement index has no correlation with the number of cases involving the litigant.

6 Discussion

The main findings of this paper are consistent with the socio-legal and empirical legal studies literature that documents that over two-thirds of disputes are settled through mediation rather than continue with trial in widely different contexts (Silbey and Merry 1986; Kaplan et al. 2008; Silveira 2017). The main mechanism of mediation discussed in these broader literature is the reduction of information asymmetries, both between plaintiff and defendants, as well as about the uncertainties of the legal process itself. While neither the existing literature nor we have information to causally examine this specific mechanism rigorously with the exception of Sadka et al. (2024), the evidence from the analysis of case-specific judgment orders is supportive of this mechanism. The contribution of this study is to examine the impact of mediation among the broader class of commercial and contractual disputes beyond labor disputes examined in Sadka et al. (2024).

In order to draw policy implications, we need to first discuss the implications of our findings through the lens of the statutory provisions within the legal system in India. These include examining the constitutionality of mediation as policy as well as identifying practical next steps such as training judges in mediation strategies.

6.1 Doctrinal Legal Analysis

The Commercial Courts Act’s primary objective is to achieve faster resolution of commercial and contractual disputes. This study demonstrates that resolution through mediation (settlement) is negatively correlated with its duration and its pending status. These findings are consistent with the legislative intent behind establishing dedicated commercial courts and emphasizing settlement. From a legal standpoint, the reduction in case duration and pendency aligns with constitutional mandates for speedy justice and the efficiency principles underlying procedural laws such as the Code of Civil Procedure, 1908.

Constitutionality of “Mandatory” Mediation A critical feature of the reform is mandatory mediation preceding a lawsuit (Section 12 A, Commercial Courts Act, 2015). Our finding of a positive effect on the profitability of plaintiff without any long-term losses for defendant firms following settlement in lawsuits underscores mediation’s role in achieving beneficial outcomes for litigants. Mandating mediation before a dispute becomes a lawsuit could potentially prevent new suits. Although this paper examines what happens when a judge recommends mediation within commercial lawsuits, it is plausible that disputes that undergo formal mediation by law could benefit both litigants and avert any costs associated with pursuing a lawsuit altogether. This motivates further research to examine the effects of mandating mediation as a dispute resolution strategy in

commercial cases under the Commercial Courts Act as well as in other legislation dealing with commerce and industry.

The findings also provide a basis for defending mediation against constitutional challenges, which argue that requiring mediation would impede access to court. The Supreme Court of India dismissed arguments against mandatory mediation before filing a lawsuit in *Patil Automation (P) Ltd. v. Rakheja Engineers (P) Ltd.*, 2022, which had challenged its constitutional validity. This paper provides empirical support by demonstrating that mediation-based settlement could actually lead to an improvement relative to the status quo.¹⁰

This paper also addresses philosophical challenges against mediation, which is premised on the fact that mediation is essentially a consent-based system, and thus, making it mandatory is contradictory to its philosophy. However, there is support from the legal literature such as Sander et al. (1996) and Sander (2007), which classify cases referred to mediation as categorical and discretionary. In categorical cases, judges typically have no discretion and mediation is automatic. Some authors (for example, Wissler 2002) have critiqued this, arguing that the benefits of mediation are not supported by empirical evidence. This paper counters these apprehensions by providing empirical evidence in support of mediation.

6.2 Random Assignment and Mediation

Since the random assignment policy in 2020, the digital Case Information System (CIS 3.2) randomly allocates newly filed lawsuits to judges to minimize forum shopping, systematic bias, and increase transparency.¹¹ We find support that this assignment system works well in practice. Specifically, we do not find any significant association between any of the judge’s characteristics, including their settlement index, and case and litigant-specific characteristics.

The finding that there exists a distribution of judge-specific settlement index - that some judges are able to settle more cases than others - indicates that effective mediation is a distinct judicial skill, not merely an incidental function. Legally, this has important implications for judicial training, performance evaluation, and potentially even judicial appointments in commercial courts. It supports the development of specialized judicial education programs focused on negotiation, conflict resolution, and mediation techniques

¹⁰From a legal standpoint, any further challenges (for example, citing an exception for requiring mediation under urgency for an interim relief) could also be resolved backed by the empirical findings of this paper. Since the law does not define what is meant by “urgent”, the results in this paper provide support for a strict reading of what is statutorily meant by “urgent”, incentivizing mediation-based settlement in most cases.

¹¹This system is part of a larger effort to introduce transparency and objectivity in the Indian judicial system at the district level. The basis of this measure is that every case should have a similar outcome irrespective of the judge on the case.

for commercial court judges.

Although judicial training academies exist in every state in India, and judicial officers are provided some form of training, specific modules facilitating mediation-based settlement are lacking in these training programs. The systemic problems in the judicial training philosophy and practice are immense and are documented extensively by the Supreme Court itself in the *All India Judges' Association Case* and in literature such as Oberoi (2018) and by Chen et al. (2026) in the judicial context outside India.

6.3 Way Forward

There are two potential policy implications of this study. First, we now have empirical evidence in the context of commercial dispute resolution in India that suggests that mediation increases the efficiency of dispute resolution process when measured in terms of case duration or pending status. The findings are also relevant when examining petitions challenging the constitutional validity of mandating mediation and serves as an example of how commercial dispute resolution codes can be designed in common law legal systems.

Second, this paper documents a wide variation in the extent of mediation abilities among commercial court judges. Looking at it from a narrow perspective, this enabled causal identification for this paper. However, this raises important questions about the judicial organizational structure in terms of training judges in the art of negotiating settlement through mediation.

7 Conclusion

This paper is among the first to examine the causal effects of court-mediated settlement on litigating firms' welfare in contract enforcement and commercial dispute litigation. Random assignment of cases to judges in commercial courts in India has enabled causal inference by introducing exogenous variation in judge "leniency" or settlement-propensity to determine the final case outcome of commercial and contractual cases. Randomization also introduces independence between any of the litigating firms' identity and judge identity, minimizing concerns of forum-shopping where litigants may game the assignment system to get a favorable judge for their case.

We find that settling a case before undergoing full trial is beneficial both for case-level metrics such as reduced duration and lower pending status as well as for litigant welfare. Whereas the profit effects are unequivocally positive for plaintiff firms, the effects on defending firms are a bit more subtle. These firms first experience a decline in their profit, which follows after the filing of their case in the court. However, the trend reverses when the defending firms settle their case through mediation.

Additional research is needed to interpret the policy implication of this finding on

the long-run welfare of litigants and the broader economy. Should the role of courts be to facilitate settlement rather than trial? If so, why couldn't the litigants have settled the dispute themselves before filing the case in the court in the first place? To some extent, this could reflect overoptimism among litigants in obtaining a decision in their favor through full trial, which a judge can correct. Indeed, Sadka et al. (2024) show that this is a possibility in the context of labor disputes in Mexican labor courts.

Another explanation is that litigants prefer formal approval to the resolution of their dispute, which is provided by courts. Mediation enables them to resolve their dispute faster *and* with a formal, state-supported mandate. This supports an interpretation where the state still plays an important role in contract enforcement even when the litigants themselves may resolve their dispute through mutual reconciliation. The state mandate could minimize future disagreements.

Court-facilitated mediation could be a potential solution to reduce the duration of legal cases in courts and reduce pending backlog. This could be a win-win if mediation generates positive welfare effects for plaintiff without negative effects for the defendants. Even in disputes such as debt recovery and bankruptcy suits where the outcome results in the liquidation of the defending firm, this need not be viewed as a "cost". In fact, a rich literature on capital misallocation suggests that many inefficient firms continue to exist instead of exiting at the cost of aggregate inefficiency (Hsieh and Klenow 2009; Chatterjee et al. 2025).

References

- Amirapu, Amrit**, “Justice Delayed Is Growth Denied: The Effect of Slow Courts on Relationship-Specific Industries in India,” *Economic Development and Cultural Change*, October 2021, 70 (1), 415–451. Publisher: The University of Chicago Press.
- Arnold, David, Will Dobbie, and Crystal S Yang**, “Racial Bias in Bail Decisions*,” *The Quarterly Journal of Economics*, November 2018, 133 (4), 1885–1932.
- Bamieh, Omar, Decio Coviello, Andrea Ichino, and Nicola Persico**, “Effect of Business Uncertainty on Turnover,” *Journal of Labor Economics*, 2025, pp. 000–000.
- Banerjee, Abhijit V. and Esther Duflo**, “Reputation Effects and the Limits of Contracting: A Study of the Indian Software Industry*,” *The Quarterly Journal of Economics*, 08 2000, 115 (3), 989–1017.
- Bhuller, Manudeep and Henrik Sigstad**, “Errors and monotonicity in judicial decision-making,” *Economics Letters*, June 2022, 215, 110486.
- Blattman, Christopher, Alexandra C. Hartman, and Robert A. Blair**, “How to Promote Order and Property Rights under Weak Rule of Law? An Experiment in Changing Dispute Resolution Behavior through Community Education,” *The American Political Science Review*, 2014, 108 (1), 100–120.
- Cai, Jing and Adam Szeidl**, “Indirect Effects of Access to Finance,” *American Economic Review*, August 2024, 114 (8), 2308–51.
- Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H. C. Sant’Anna**, “Difference-in-differences with a Continuous Treatment,” February 2024.
- Chatterjee, Shoumitro, Kala Krishna, Kalyani Padmakumar, and Yingyan Zhao**, “No Country for Dying Firms: Evidence from India,” Working Paper 33830, National Bureau of Economic Research May 2025.
- Chemin, Matthieu**, “Does Court Speed Shape Economic Activity? Evidence from a Court Reform in India,” *The Journal of Law, Economics, and Organization*, 11 2010, 28 (3), 460–485.
- , “Judicial Efficiency and Firm Productivity: Evidence from a World Database of Judicial Reforms,” *The Review of Economics and Statistics*, March 2020, 102 (1), 49–64.
- , **Paul Kimalu, and Simon Newman-Bachand**, “Courts, Crime and Economic Performance: Evidence from a Judicial Reform in Kenya,” *Journal of Public Economics*, 2024, 231, 105035.
- Chen, Daniel, Manuel Ramos-Maqueda, and Bernardo Silveira**, “Bureaucratic Training and State Capacity: Experimental Evidence from Peru’s Judiciary,” Working Paper, National Bureau of Economic Research April 2026.

- Coviello, Decio, Andrea Ichino, and Nicola Persico**, “THE INEFFICIENCY OF WORKER TIME USE,” *Journal of the European Economic Association*, 2015, 13 (5), 906–947.
- , **Luigi Moretti, Giancarlo Spagnolo, and Paola Valbonesi**, “Court Efficiency and Procurement Performance,” *The Scandinavian Journal of Economics*, 2018, 120 (3), 826–858.
_eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/sjoe.12225>.
- Djankov, Simeon, Oliver Hart, Caralee McLiesh, and Andrei Shleifer**, “Debt Enforcement around the World,” *Journal of Political Economy*, 2008, 116 (6), 1105–1149.
- , **Rafael La Porta, Florencio Lopez de Silanes, and Andrei Shleifer**, “Courts*,” *The Quarterly Journal of Economics*, May 2003, 118 (2), 453–517.
- Faulkender, Michael and Rong Wang**, “Corporate Financial Policy and the Value of Cash,” *The Journal of Finance*, 2006, 61 (4), 1957–1990.
- Frandsen, Brigham, Lars Lefgren, and Emily Leslie**, “Judging Judge Fixed Effects,” *American Economic Review*, January 2023, 113 (1), 253–277.
- Ghosh, Shubhashis**, “Commercial Dispute Resolution in India: Assessing the Efficacy of Commercial Courts Act,” *National Law School of India Review*, 2018, 30 (1), 15–34.
- Hartman, Alexandra C., Robert A. Blair, and Christopher Blattman**, “Engineering Informal Institutions: Long-Run Impacts of Alternative Dispute Resolution on Violence and Property Rights in Liberia,” *The Journal of Politics*, 2021, 83 (1), 381–389.
- Hsieh, Chang-Tai and Peter J. Klenow**, “Misallocation and Manufacturing TFP in China and India,” *The Quarterly Journal of Economics*, November 2009, 124 (4), 1403–1448.
- Kaplan, David S., Joyce Sadka, and Jorge Luis Silva-Mendez**, “Litigation and Settlement: New Evidence from Labor Courts in Mexico,” *Journal of Empirical Legal Studies*, 2008, 5 (2), 309–350.
- Kling, Jeffrey R.**, “Incarceration Length, Employment, and Earnings,” *American Economic Review*, June 2006, 96 (3), 863–876.
- Kondylis, Florence and Mattea Stein**, “The Speed of Justice,” *The Review of Economics and Statistics*, May 2023, 105 (3), 596–613.
- Liu, Ernest, Yi Lu, Wenwei Peng, and Shaoda Wang**, “Judicial Independence, Local Protectionism, and Economic Integration: Evidence from China,” September 2022.
- Mattsson, Martin and Ahmed Mushfiq Mobarak**, “Formalizing Dispute Resolution: Effects of Village Courts in Bangladesh,” June 2025.
- Nunn, Nathan**, “Relationship-Specificity, Incomplete Contracts, and the Pattern of Trade*,” *The Quarterly Journal of Economics*, May 2007, 122 (2), 569–600.

- Oberoi, Geeta**, “Limitations of Induction Trainings Offered to Magistrates by State Judicial Educators in India,” *Athens Journal of Law*, 2018, 4 (4), 301–316.
- Ponticelli, Jacopo and Leonardo S. Alencar**, “Court Enforcement, Bank Loans, and Firm Investment: Evidence from a Bankruptcy Reform in Brazil *,” *The Quarterly Journal of Economics*, August 2016, 131 (3), 1365–1413.
- Rao, Manaswini**, “Front-line Courts As State Capacity: Evidence From India,” June 2024.
- Ryan, Nicholas**, “Contract Enforcement and Productive Efficiency: Evidence From the Bidding and Renegotiation of Power Contracts in India,” *Econometrica*, 2020, 88 (2), 383–424.
- Sadka, Joyce, Enrique Seira, and Christopher Woodruff**, “Information and Bargaining through Agents: Experimental Evidence from Mexico’s Labour Courts,” *The Review of Economic Studies*, November 2024, 91 (6), 3677–3711.
- Sandefur, Justin and Bilal Siddiqi**, “Delivering Justice to the Poor: Theory and Experimental Evidence from Liberia,” *Working Paper*, 2015.
- Sander, Frank E. A.**, “Another View of Mandatory Mediation,” *Dispute Resolution Magazine*, Winter 2007, 13 (2), 16.
- , **H. William Allen, and Debra Hensler**, “Judicial (Mis)use of ADR? A Debate,” *University of Toledo Law Review*, 1996, 27 (4), 885–886.
- Shavell, Steven**, “Suit, Settlement, and Trial: A Theoretical Analysis under Alternative Methods for the Allocation of Legal Costs,” *The Journal of Legal Studies*, 1982, 11 (1), 55–81.
- , “Settlement of Litigation May Increase or Decrease Deterrence and Social Welfare,” Working Paper 34878, National Bureau of Economic Research February 2026.
- Sigstad, Henrik**, “Monotonicity among Judges: Evidence from Judicial Panels and Consequences for Judge IV Designs,” August 2023.
- Silbey, Susan S.**, “Ideology, Power, and Justice,” in Bryant G. Garth and Austin Sarat, eds., *Justice and Power in Sociolegal Studies*, Vol. 1 of *Fundamental Issues in Law and Society Research*, Northwestern University Press, 1998.
- and **Sally E. Merry**, “Mediator Settlement Strategies,” *Law & Policy*, 1986, 8 (1), 7–32.
- Silveira, Bernardo S.**, “Bargaining With Asymmetric Information: An Empirical Study of Plea Negotiations,” *Econometrica*, 2017, 85 (2), 419–452.
- The Economic Times**, “From 1,095 days to 306: Govt data shows quicker disposal of commercial disputes,” *Economic Times* (online) 2021. PTI report dated 22 August 2021.

Visaria, Sujata, “Legal Reform and Loan Repayment: The Microeconomic Impact of Debt Recovery Tribunals in India,” *American Economic Journal: Applied Economics*, July 2009, *1* (3), 59–81.

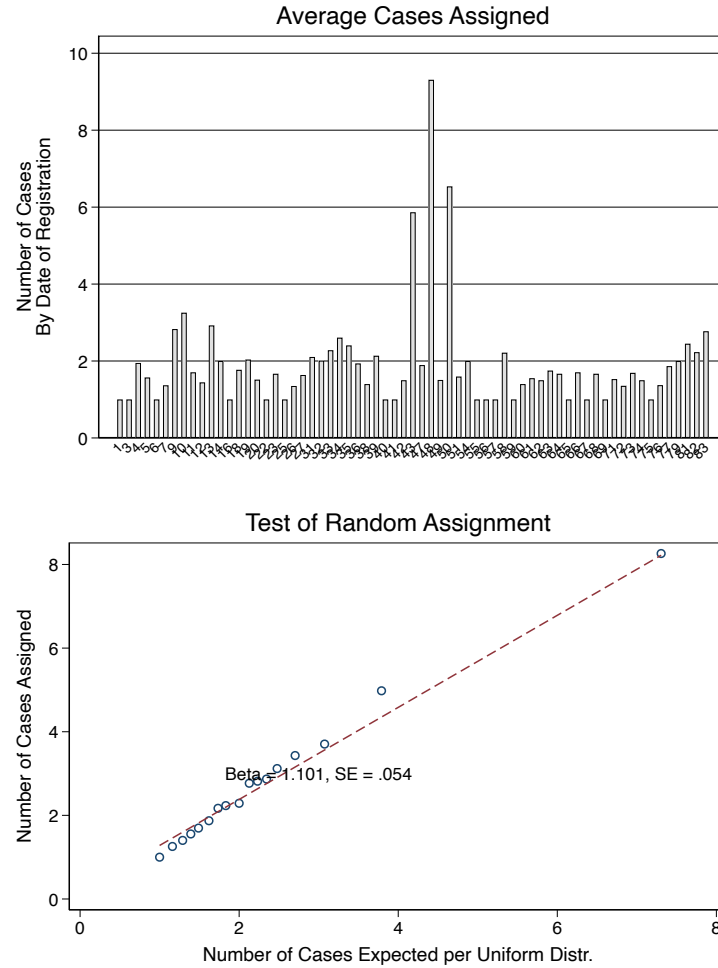
von Lilienfeld-Toal, Ulf, Dilip Mookherjee, and Sujata Visaria, “The Distributive Impact of Reforms in Credit Enforcement: Evidence From Indian Debt Recovery Tribunals,” *Econometrica*, 2012, *80* (2), 497–558. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA9038>.

Wissler, Roselle L., “Court-Connected Mediation in General Civil Cases: What We Know from Empirical Research,” *Ohio State Journal on Dispute Resolution*, 2002, *17* (3), 641–704.

—, “The effectiveness of court-connected dispute resolution in civil cases,” *Conflict Resolution Quarterly*, 2004, *22* (1-2), 55–88.

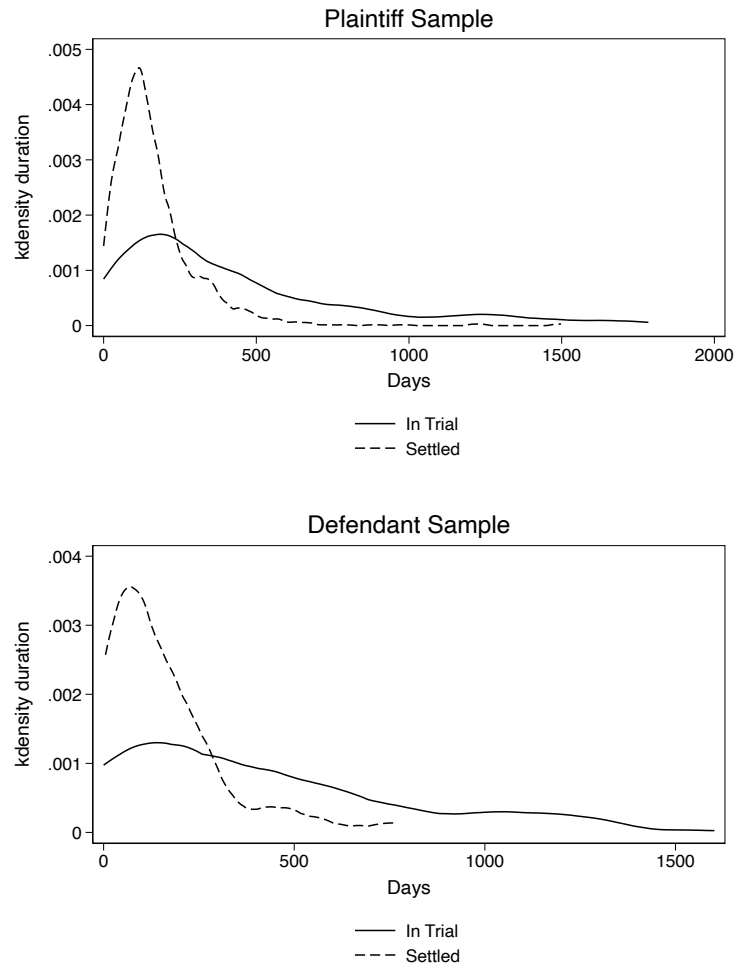
8 Figures

Figure 1: Judge Workflow Generated by Random Assignment



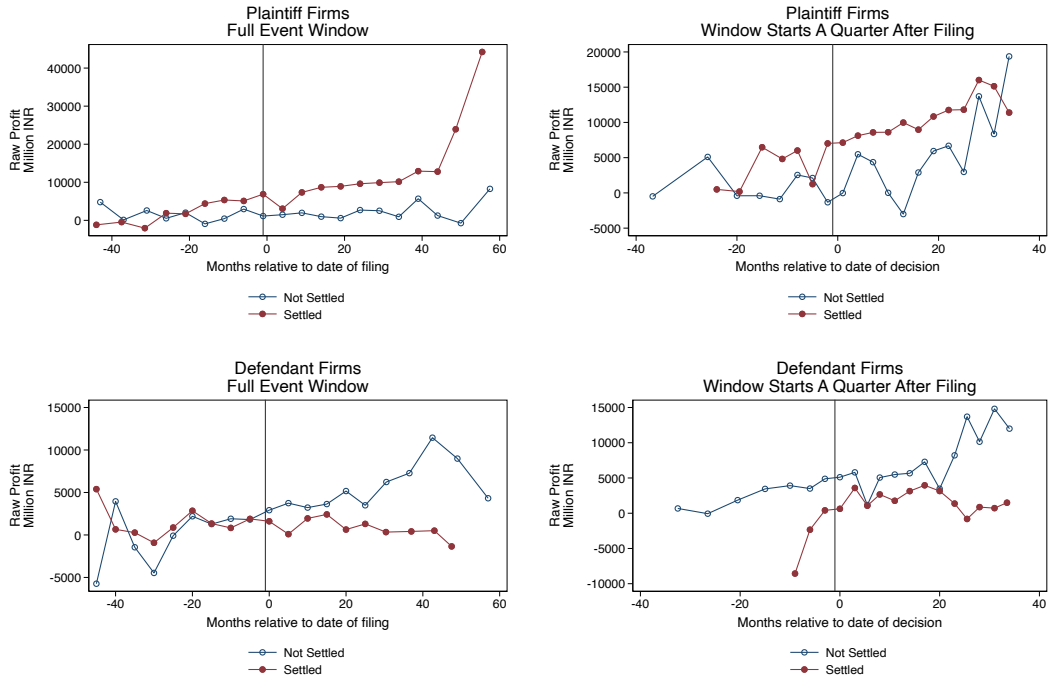
Notes: The figures above depict the number of cases assigned per date of suit registration to judges (judge ID as x-axis label in top panel) in each commercial court, respectively. The “spikes” observed in number of cases per judge ID in the histogram is due to newly added judges to the commercial courts who are randomly assigned a whole docket at the time of joining. The figure below plots the correlation between assigned (realized) workload and expected workload following a uniform random assignment algorithm on any given day.

Figure 2: Case Duration By Settlement Status



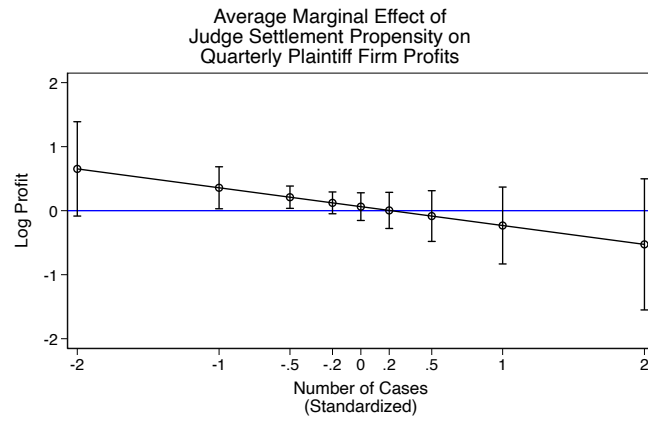
Notes: Distribution of case duration by case settlement status during mediation plaintiff (top) and defendant samples (bottom). The median duration is 132 days if settled relative to a median of 313 days among those that fail settlement in the plaintiff dataset. The numbers are similar in defendant dataset, with the mediation duration 109 days among settled cases and 297 days among those that fail to settle.

Figure 3: Raw Means of Firm-level Quarterly Profit Around Case Decision

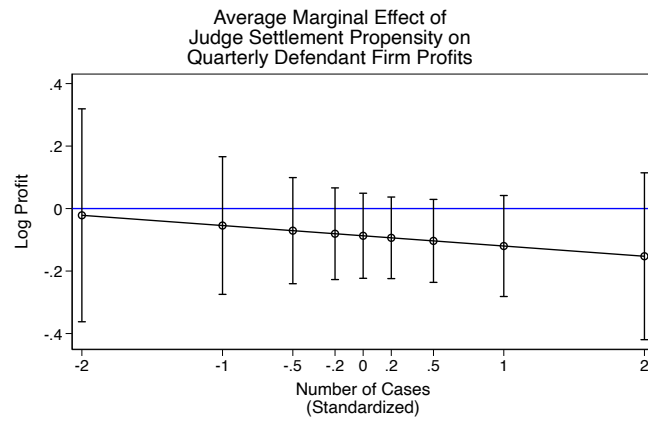


Notes: The figures above present quarterly profit in levels after deflating for inflation, measured in million INR, separately for the plaintiff and defendant samples. The groups include those that settled their suit during mediation and those that did not settle and underwent the trial process. Vertical lines denote the event time reference (1 quarter prior to filing or decision, respectively). The event time is relative to the date of filing (left) or decision (right) of a case in the respective commercial court. In our causal analysis, we examine the effect of being assigned a settlement-prone judge based on random assigned to a case since whether a case is settled or not is potentially endogenous.

Figure 4: Average Marginal Effects on Firm Profit
Panel A: Plaintiff Firms



Panel B: Defendant Firms



Notes: The figures above present average marginal effects by the standard deviation of firms' litigation intensity (number of cases) in the commercial courts, separately for plaintiff (Panel A) and defendant firm samples (Panel B). The dependent variable is log profit. The marginal effects are computed using margins and marginsplot following a heterogeneity by standardized number of cases regression.

9 Tables

Table 1: Summary Statistics and Sample Composition

Panel A: Distribution of Litigant-Type Pairs			
	(1)	(2)	(3)
	Full Case Sample	Plaintiff	Defendant
Bank–Bank	0.05	0.20	–
Bank–Firm	16.89	46.95	2.35
Bank–Individual	16.65	33.82	–
Firm–Bank	1.37	0.20	10.59
Firm–Firm	33.45	12.86	55.29
Firm–Individual	13.99	5.96	–
Individual–Bank	1.28	–	10.59
Individual–Firm	10.52	–	21.18
Individual–Individual	5.78	–	–
Cases	6,624	1,493	255

Panel B: Firm Characteristics Summary			
	(1)	(2)	(3)
	Full Prowess Sample	Plaintiff	Defendant
Age	30.165 (16.495)	37.513 (28.385)	32.772 (21.910)
Manufacturing	0.373 (0.484)	0.175 (0.381)	0.153 (0.362)
Trade and Retail	0.186 (0.389)	0.175 (0.381)	0.113 (0.318)
Services	0.166 (0.372)	0.182 (0.387)	0.129 (0.337)
Publicly Listed	0.544 (0.498)	0.617 (0.488)	0.669 (0.472)
Observations (Firms)	40,786	154	124

Notes: Panel A reports percentages of litigant-type pairs in the case-level data. Col 1 presents the distribution across the universe of cases from the two courts in our study. Col 2 presents the distribution among the subset of these cases in the plaintiff firm-matched data. Col 3 presents the distribution among the subset of defendant firm-matched data. Panel B presents the characteristics of firms in the analysis samples relative to those in the full Prowess sample. Col 1 represents the distribution of characteristics across the full sample in Prowess database. Cols 2 and 3 present the distribution of these characteristics among firms matched with the case-level data, depending on whether the firm appears as a plaintiff or as a defendant, respectively.

Table 2: Balance Table

Dep Var	Judge ID			Leave-out Settled	
	(1) All	(2) Plaintiff	(3) Defendants	(4) Plaintiff	(5) Defendants
Bank–Bank	6.55 (18.11)	13.10 (21.86)	. [†] (.)	-0.14 (0.13)	. [†] (.)
Bank–Firm	-1.37 (2.38)	3.35 (5.32)	-15.45 (11.79)	-0.01 (0.03)	-0.19* (0.09)
Bank–Individual	-1.84 (2.63)	3.02 (4.98)	–	0.00 (0.03)	–
Firm–Bank	1.74 (2.84)	24.38*** (7.81)	8.07 (11.96)	-0.18* (0.10)	0.12** (0.05)
Firm–Individual	-3.48*** (1.21)	-2.60 (4.64)	–	0.02 (0.03)	–
Individual–Bank	-2.31 (3.76)	–	-14.32* (7.61)	–	0.06 (0.06)
Individual–Firm	-2.50 (2.09)	–	-7.88 (7.41)	–	-0.01 (0.03)
Individual–Individual	-3.32 (2.38)	–	–	–	–
Age	–	0.00 (0.03)	0.02 (0.08)	-0.00** (0.00)	0.00 (0.00)
Non-Finance Firm	–	4.86 (3.23)	-3.68 (5.94)	-0.06 (0.04)	0.00 (0.04)
Bank	–	-4.26* (2.41)	-6.85 (12.37)	0.04* (0.02)	-0.12* (0.06)
Observations	6,081	1,489	247	1,450	216
F-stat	1.18	3.64	1.92	0.96	1.12
Joint p-value	0.326	0.003	0.095	0.485	0.380
City Fixed Effects	✓	✓	✓	✓	✓
Registration-Month FE	✓	✓	✓	✓	✓
Case Type FE	✓	✓	✓	✓	✓

Notes: Robust standard errors in parentheses. The dependent variable in Columns (1)–(3) is `judge_id`; in Columns (4)–(5) it is the leave-out mean of settlement. All regressions include fixed effects for the city, registration month-year, and commercial case type (execution, appeals, or original suit). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, [†] omitted due to no variation.

Table 3: Probability of Settlement by Settlement-Propensity of Assigned Judge

	(1) Case Settled Plaintiff	(2) Case Settled Plaintiff	(3) Case Settled Defendant	(4) Case Settled Defendant
Leave-out Settled (SD)	0.177*** (0.0132)	0.148*** (0.0216)	0.216*** (0.0341)	0.233*** (0.0318)
Financial Firm x Leave-out Settled (SD)		0.0251 (0.0184)		-0.0666** (0.0284)
Financial Firm		0.185*** (0.0477)		-0.0175 (0.0787)
Observations	1470	1470	235	235
No. Judges	30	30	30	30
City FE	Y	Y	Y	Y
Registration-Month FE	Y	Y	Y	Y
Case-Type FE	Y	Y	Y	Y
Control Mean	0.69	0.69	0.0	0.0
Control SD	0.47	0.47	0.0	0.0
Adj R-Squared	0.368	0.381	0.384	0.382
F-stat	178.1	114.9	40.30	34.52

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: This table presents the “first stage” examining the likelihood that a case assigned to a settlement-prone judge is settled. Columns 1 and 2 use matched-firm case-level data from the plaintiff dataset whereas columns 3 and 4 use matched-firm case-level data from the defendant dataset. The main explanatory variable is the judge settlement propensity constructed as leave-out average of settlement rates in cases other than the specific firm in normalized units (standard deviation from the average settlement propensity). Since cases were randomly assigned, we interpret the coefficients in this table as causal effect parameters. Columns 1 and 3 examine the effect of being assigned a settlement-prone judge on whether a case is settled through mediation (coded 1) or not (coded 0). Columns 2 and 4 present heterogeneity analysis based on whether the litigating firm is a financial sector firm or not. All specifications include randomization strata fixed effects (city specific court and case allocation batch fixed effects) in addition to controlling for specific type of the case. Standard errors are clustered at the judge-level. Control mean and standard deviation are calculated for cases assigned to judges within 0.1 standard deviation of the average settlement rate. While many plaintiffs (69%) assigned to judge within 0.1 standard deviation of the average settlement rate settle their cases, no defendants in this group settle. Thus, both control mean and standard deviation are 0 in Col 3-4.

Table 4: Settlement and Efficiency of Dispute Resolution

	(1) Pending Plaintiff	(2) Duration (Days) Plaintiff	(3) Pending Defendant	(4) Duration (Days) Defendant
Panel A: OLS				
Case Settled	-0.163** (0.0598)	-50.98** (19.23)	-0.234** (0.108)	-136.9*** (36.81)
Control Mean	0.126	437.7	0.0865	406.8
Control SD	0.333	399.9	0.282	371.4
Adj R-Squared	0.198	0.701	0.217	0.779
Panel B: Judge Settlement Index				
Leave-out Settled (SD)	-0.0841** (0.0395)	-4.257 (17.26)	-0.202** (0.0881)	-26.02 (37.26)
Observations	1470	1420	235	216
No. Judges	30	30	30	30
City FE	Y	Y	Y	Y
Registration-Month FE	Y	Y	Y	Y
Case-Type FE	Y	Y	Y	Y
Control Mean	0	248.75	0.23	278
Control SD	0	228.2	0.44	197.96
Adj R-Squared	0.188	0.696	0.315	0.763

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: This table examines the association between mediation-based settlement (Panel A) and the causal effects of being assigned a settlement-prone judge (Panel B) on some of the efficiency metrics of dispute resolution - pending status and suit duration. Columns 1 and 2 use matched-firm case-level data from the plaintiff dataset whereas columns 3 and 4 use matched-firm case-level data from the defendant dataset. The number of observations slightly differ between odd and even columns because of missing duration data for cases that are pending at the end of the study period. The main explanatory variable in Panel A is whether a particular case is settled through mediation or not. Note that this could be endogenous and thus, the coefficients presented in this table should only be interpreted as correlational. The main explanatory variable in Panel B is the judge settlement propensity constructed as leave-out average of settlement rates in cases other than the specific firm. Since cases were randomly assigned, we interpret the coefficients in this table as causal effect parameters. All specifications include randomization strata fixed effects (city specific court and case allocation batch fixed effects) in addition to controlling for specific type of the case. Standard errors are clustered at the judge-level. Control mean and standard deviation are calculated for cases assigned to judges within 0.1 standard deviation of the average settlement rate. Since many plaintiffs assigned to judge within 0.1 standard deviation of the average settlement rate settle their cases, their pending status is 0. Among defendants, 23% cases assigned to judge with average settlement rate continue to be pending.

Table 5: Settlement and Efficiency of Dispute Resolution: By Financial Sector Heterogeneity

	(1) Pending Plaintiff	(2) Duration (Days) Plaintiff	(3) Pending Defendant	(4) Duration (Days) Defendant
Financial Firm x Leave-out Settled (SD)	-0.00496 (0.0193)	-41.24* (20.90)	-0.0112 (0.0247)	28.01 (23.60)
Leave-out Settled (SD)	-0.0807** (0.0354)	29.09 (23.84)	-0.199** (0.0861)	-32.26 (37.53)
Financial Firm	0.00669 (0.0322)	-37.91 (27.71)	-0.0321 (0.0292)	-46.92 (33.45)
Observations	1470	1420	235	216
No. Judges	30	30	30	30
City FE	Y	Y	Y	Y
Registration-Month FE	Y	Y	Y	Y
Case-Type FE	Y	Y	Y	Y
Adj R-Squared	0.187	0.701	0.310	0.763

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: This table examines heterogeneity in Table 4 by whether the litigating firm is a financial firm or not. Columns 1 and 2 use matched-firm case-level data from the plaintiff dataset whereas columns 3 and 4 use matched-firm case-level data from the defendant dataset. Financial firms include banks and other financial service providers such as fintec and non-banking financial companies (NBFCs). The number of observations slightly differ between odd and even columns because of missing duration data for cases that are pending at the end of the study period. All specifications include randomization strata fixed effects (city specific court and case allocation batch fixed effects) in addition to controlling for specific type of the case. Standard errors are clustered at the judge-level.

Table 6: Reduced Form Effects on Firms' Profitability: Around Filing and Around Decision

	(1) Log Profit Plaintiff Around Filing	(2) Log Profit Defendants Around Filing	(3) Log Profit Plaintiff All Periods	(4) Log Profit Defendants All Periods
Post Filing	-0.0455 (0.121)	0.0593 (0.185)	-0.0335 (0.125)	-0.00116 (0.191)
Post Filing $\times$ Leave-out Settled (SD)	0.179** (0.0636) { 0.01 }	-0.294** (0.115) { 0.127 }	0.183** (0.0707) { 0.044 }	-0.311** (0.112) { 0.035 }
Post Decision			0.0390 (0.150)	0.0465 (0.142)
Post Decision $\times$ Leave-out Settled (SD)			0.0420 (0.0805) { 0.61 }	0.164 (0.117) { 0.286 }
Observations	1819	1260	2265	1539
No. Judges	19	21	19	21
No. Firms	70	56	71	56
City FE	Y	Y	Y	Y
Registration-Month FE	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y
Mean Dep Var	6434.2	5795.8	6434.2	5795.8
SD Dep Var	30894.3	20867.0	30894.3	20867.0
Adj R-Squared	0.834	0.846	0.836	0.846

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: Sample includes firm-level quarterly profit among firms matched to cases that were randomly assigned to judges in the sample commercial courts. The time period includes quarters from 2011 to 2024, with “Post Filing” dummy denoting quarters following the filing date of the first settled case and “Post Decision” period denoting quarters after the decision date of that case. The time period in Col 1-2 is surrounding the timing of filing the suit but excluding quarters just before its resolution. Col 3-4 include the full panel. Mean and standard deviation of profit levels are reported in the footer. Standard errors are clustered by judge. Bootstrapped p-values in {.}.

Table 7: Real Economic Outcomes: Wage Bill

	(1) Log Wage Bill Plaintiff Around Filing	(2) Log Wage Bill Defendants Around Filing	(3) Log Wage Bill Plaintiff All Periods	(4) Log Wage Bill Defendants All Periods
Post Filing	-0.298 (0.205)	-0.265 (0.223)	-0.198 (0.161)	-0.189 (0.160)
Post Filing $\times$ Leave-out Settled (SD)	-0.0619 (0.101) { 0.597 }	0.0295 (0.185) { 0.893 }	-0.00389 (0.0838) { 0.968 }	-0.0269 (0.176) { 0.860 }
Post Decision			0.0625 (0.144)	-0.389** (0.150)
Post Decision $\times$ Leave-out Settled (SD)			0.153** (0.0708) { 0.034 }	-0.0724 (0.0840) { 0.322 }
Observations	2363	1661	2910	2051
No. Judges	19	21	19	21
No. Firms	70	56	71	56
City FE	Y	Y	Y	Y
Registration-Month FE	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y
Mean Dep Var	6193.4	8631.4	6193.4	8631.4
SD Dep Var	8392.7	25564.3	8392.7	25564.3
Adj R-Squared	0.935	0.939	0.928	0.920

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: Sample includes firm-level quarterly wage bill among firms matched to cases that were randomly assigned to judges in the sample commercial courts. The time period includes quarters from 2011 to 2024. “Post Filing” dummy denotes quarters following the filing date of the first settled case. “Post Decision” period denotes quarters after the decision date of that case. All columns include the full panel. Mean and standard deviation of wage-bill levels reported in the footer. Standard errors are clustered by judge. Bootstrapped p-values in {.}.

Table 8: Real Economic Outcomes: Long-run Investment

	(1) Log Investments Plaintiff Around Filing	(2) Log Investments Defendants Around Filing	(3) Log Investments Plaintiff All Periods	(4) Log Investments Defendants All Periods
Post Filing	-0.0960 (0.0604)	0.0288 (0.105)	-0.0464 (0.0618)	0.00402 (0.0839)
Post Filing $\times$ Leave-out Settled (SD)	-0.0223 (0.0657) { 0.722 }	0.0210 (0.0603) { 0.829 }	0.0219 (0.0599) { 0.728 }	-0.0168 (0.0551) { 0.809 }
Post Decision			-0.0453 (0.0708)	0.0585 (0.0962)
Post Decision $\times$ Leave-out Settled (SD)			0.0505* (0.0251) { 0.131 }	-0.0209 (0.0448) { 0.705 }
Observations	2356	1651	2894	2041
No. Judges	19	20	19	20
City FE	Y	Y	Y	Y
Registration-Month FE	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y
Mean Dep Var	187418.1	779964.4	187418.1	779964.4
SD Dep Var	251535.0	2643113.8	251535.0	2643113.8
Adj R-Squared	0.965	0.952	0.968	0.956

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: Sample includes firm-level quarterly investments in long-run assets among firms matched to cases that were randomly assigned to judges in the sample commercial courts. The time period includes quarters from 2011 to 2024. “Post Filing” dummy denotes quarters following the filing date of the first settled case. “Post Decision” period denotes quarters after the decision date of that case. All columns include the full panel. Mean and standard deviation of investment levels reported in the footer. Standard errors are clustered by judge. Bootstrapped p-values in {.}.

Table 9: Specifics of Settlement

	(1)	(2)	(3)	(4)
	Ex-Parte Order	No. Pages Order	Decree Amount Missing	Decree Amount (100,000 INR)
Leave-out Settled (SD)	0.0571*** (0.0148)	1.468** (0.682)	-0.0439 (0.0273)	0.893* (0.438)
Observations	1091	1091	1091	471
No. Judges	24	24	24	17
City FE	Y	Y	Y	Y
Registration-Month FE	Y	Y	Y	Y
Case-Type FE	Y	Y	Y	Y
Control Mean	0.387	16.06	0.742	6.761
Control SD	0.495	9.487	0.445	6.311
Adj R-Squared	0.302	0.237	0.390	0.995

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The variables in this table are generated from the text of the orders of the case-level sample. Since orders are generated once a decision had been made, the sample in this table is the subset of cases with a decision. We used Claude Opus 4.6 model to read all textual information from the final orders scraped from court websites as PDF documents. Based on the textual data in these orders, the AI model generated outcome variables to illustrate whether defendant was absent at the time of the order (Col 1 Ex-Parte Order), number of pages in the order (Col 2), and decree amount (Col 3-4; including whether this information is missing in the case order). Standard errors are clustered by judge.

Table 10: Evidence from Recoveries

	(1) Recoveries (Million INR) Plaintiff	(2) Log Recoveries Plaintiff	(3) Recoveries (Million INR) Defendant	(4) Log Recoveries Defendant
Post Filing	-184.0* (100.2)	-0.116 (0.0922)	-280.5** (127.4)	-0.295* (0.144)
Post Filing $\times$ Leave-out Settled (SD)	-10.10 (45.15)	-0.00627 (0.0318)	27.76 (44.52)	-0.0355 (0.0501)
Post Filing $\times$ Financial Firms	147.8** (62.91)	-0.0300 (0.146)	64.42 (59.02)	-0.0563 (0.111)
Post Filing $\times$ Financial Firms $\times$ Leave-out Settled (SD)	-148.3* (82.54)	-0.540*** (0.126)	-100.3 (60.08)	-0.221** (0.103)
Post Decision	-36.60 (84.06)	0.0159 (0.0790)	-42.66 (58.02)	-0.0718 (0.0881)
Post Decision $\times$ Leave-out Settled (SD)	-103.3 (77.15)	0.00529 (0.0288)	-15.13 (44.20)	0.00737 (0.0542)
Post Decision $\times$ Financial Firms	-120.4 (124.7)	-0.000277 (0.134)	10.06 (62.73)	0.117 (0.125)
Post Decision $\times$ Financial Firms $\times$ Leave-out Settled (SD)	320.0** (118.6)	0.864*** (0.194)	85.34 (66.03)	0.305** (0.119)
Observations	1394	1394	765	765
No. Judges	16	16	15	15
City FE	Y	Y	Y	Y
Registration-Month FE	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y
Adj R-Squared	0.784	0.950	0.954	0.938

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: Recovery is measured using “write back of provisions” reported on firms’ quarterly balance sheets. In Prowess, this variable is defined as “financial indicator that tracks funds previously set aside for expected liabilities or losses (like bad debts or taxes) that were not ultimately required and were consequently reversed into the profit and loss statement”. While for non-financial firms, provisions written back could arise from tax provisions or depreciation provisions, for financial firms like banks, this is mainly provisions written back from bad debts.

Online Appendix

B Data and Variable Construction

B.1 Data Sources and Preparation

The analysis combines three primary data sources: (1) case records from Delhi and Karnataka courts, (2) judicial assignment records, and (3) corporate financial statements from the Prowess dx database. We implemented a multi-stage cleaning and merging process to construct the final analytic dataset.

B.2 Case Record Processing

The raw case data (`Delhi_Kar_CC`) underwent extensive cleaning to standardize party names and case outcomes. We removed non-alphabetic characters, standardized legal entity suffixes (e.g., converting "LIMITED" to "LTD"), and created indicators for corporate parties using a dictionary of common business terms supplemented by manual review. Case dispositions were categorized into six mutually exclusive outcomes: ALLOWED, CONTESTED, DISMISSED, ORDERED, SET ASIDE, and SETTLED, with spelling variations normalized through automated and manual corrections.

B.3 Judge Data Merging

We matched cases to judges using court identifiers and hearing dates, ensuring temporal alignment with judicial tenures. The merge accounted for judge transfers between courts by verifying assignment periods against official records. For Karnataka courts, we implemented a court-by-court matching procedure (courts L32-L39) before combining results, while Delhi cases were appended after processing. This yielded a judge-case panel covering all observed dispositions.

B.4 Firm Identification and Matching

Corporate parties were identified using a combination of:

- Regular expressions for common business suffixes
- Manual review of entity names
- Fuzzy string matching (Jaro-Winkler similarity > 0.9) with corporate registries

The matching process successfully linked 476 unique firms (249 petitioners and 227 respondents) to their financial records. We verified matches through manual checks of a random sample (10% of matched cases), finding 98% accuracy in entity identification.

B.5 Financial Data Integration

Balance sheet information was merged using unique company identifiers (`co_code`), with quarterly financials aligned to case timing. We retained only exact matches between legal records and financial data, yielding eight complete datasets (petitioners/respondents $\times$ income/expenses/assets/capital). Financial variables were inflation-adjusted using RBI price indices and winsorized at the 1st/99th percentiles to mitigate outlier effects.

B.6 Case Order and Judgment Data Extraction

We downloaded pdf files on case orders and final judgments associated with each case number in our final analysis sample using case-specific unique id from the respective court websites manually by following the steps below:

1. Go to ecourts webpage:
 - (a) Bengaluru: <https://bengaluru.dcourts.gov.in/court-orders-search-by-court-number/>
 - (b) Delhi: <https://delhidistrictcourts.nic.in/casestatus>
2. For Delhi, you have the following options to select the case details:
 - (a) For each court location, click on Case orders > View Orders. This will take you to a different webpage
 - (b) Click on: Search by case number
 - (c) Enter the case number from the analysis dataset to search for final orders for the case number
 - (d) Download and save the document under “final order” and save using the case number as the file name in the project folder
3. For Bengaluru, you will need to select the following from each of the drop-downs:
 - (a) Court Complex: COMMERCIAL COURT COMPLEX
 - (b) Case type: use the value unde ‘case_num’ variable in the analysis dataset. For example: Com.A.A./113/2020, Com.A.A is the case type
 - (c) Case number: For example, in Com.A.A./113/2020, 113 is the case number
 - (d) Year: For example, in Com.A.A./113/2020, 2020 is the year
 - (e) Enter captcha
 - (f) Download the document under “final order” and save using the format as in the example ‘AA1132020_final’ in the project folder
 - (g) If there are multiple documents: check the date and download latest one. Sometimes, the document is labeled ‘Judgment’

Next, we used Claude Opus 4.6 that generates a python script to automate the extraction of structured legal and economic variables from PDF files of the commercial court orders. Utilizing the ‘pdfminer’ library, the script converts PDF documents into raw text, which is subsequently parsed using regular expressions (regex) and keyword-based heuristics. To support large datasets efficiently, the script operates incrementally by default following the manual download process occurring in phases.

The algorithm isolates over 30 distinct data points per document, mapping them into the following core categories:

Case Metadata & Parties: Captures the case number, order date, institution date, court name, and presiding judge using localized text patterns. It extracts plaintiff and defendant names by targeting preamble structures and cleans the output by removing parenthetical advocate details. The court’s city (e.g., Bengaluru, Delhi) is derived from text mentions or file paths.

Dispute Characteristics: The script infers the case type (e.g., Commercial Suit, Arbitration Petition) and order type (Final, Interim, Issue Framing) primarily from file naming conventions. The script checks the text against predefined keyword dictionaries to categorize the underlying nature of the suit (e.g., “Specific Performance”, “Recovery of Money”) and the relevant industry sector (e.g., “Banking & Finance”).

Economic Variables: The script uses a dedicated regex pattern to identify all Indian Rupee amounts, automatically converting text units like “lakhs” and “crores” into standard float values. It assigns these figures to specific variables—such as contract value, claim amounts, decree amounts, and bank guarantees—by searching for the largest monetary value within a 300-character window of specific trigger keywords. Percentage rates for interest and liquidated damages are also isolated.

Legal Outcomes & Flags: The script determines the final disposition (e.g., “Decreed”, “Dismissed”, “Settled”) by applying tiered priority regex searches strictly to the final 20% to 50% of the document, ensuring conditional language earlier in the text does not override the actual judgment. Binary flags capture the presence of interim relief, ex-parte proceedings, arbitration clauses, cited precedents, legal provisions, and mentions of the COVID-19 pandemic.

C Appendix Figures

Figure A1: Random Allocation

CASES AVAILABLE FOR RANDOM ALLOCATION

Random Allocation

*Date of Allocation: 25-02-2020

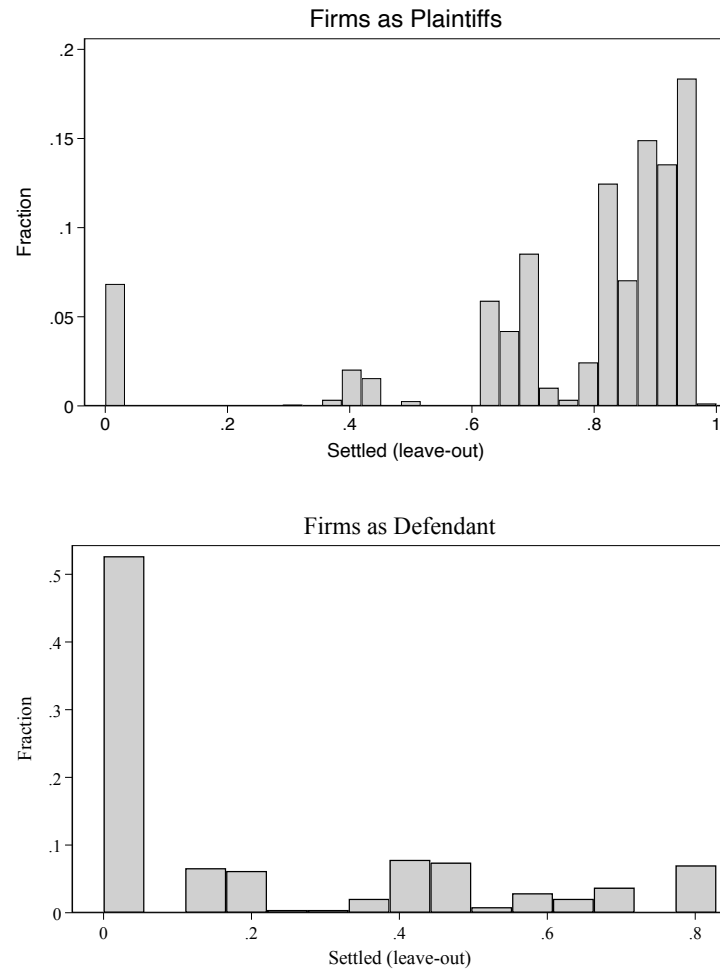
Note: Red color indicates that, The Case Type is not assigned to Court

	☒ Select All	Party Name
Total Cases : 2		
1	☒ CS (COMM)/743/2020	test VS srit
2	☒ CS (COMM)/745/2020	Rontan Singh VS Swaraj kaushal

Submit

Notes: Screenshot from eCourtIS 3.0

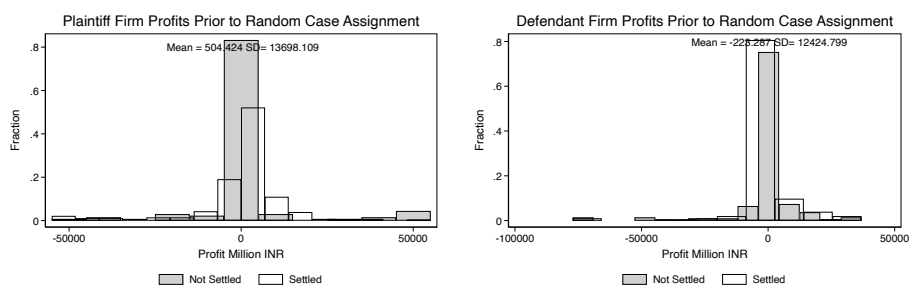
Figure A2: Settlement Propensity by Assigned Judges



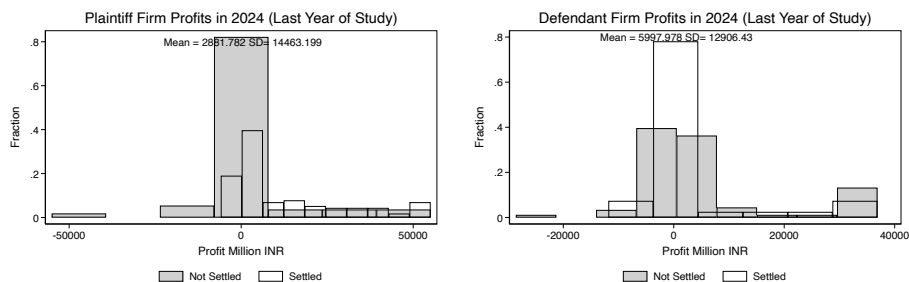
Notes: The figures show the distribution of the leave-out settlement average by judge based on cases for the plaintiff sample (top) and the defendant sample (bottom). The leave-out settlement rate of judges in the plaintiff data has a mean of 0.76 with 0.24 standard deviation whereas in the defendant dataset, the mean and standard deviation are 0.22 and 0.27, respectively.

Figure A3: Profit Distribution by Litigant Type Prior to Random Case Assignment Policy and at the End of the Study Period

Panel A: Prior to Random Assignment

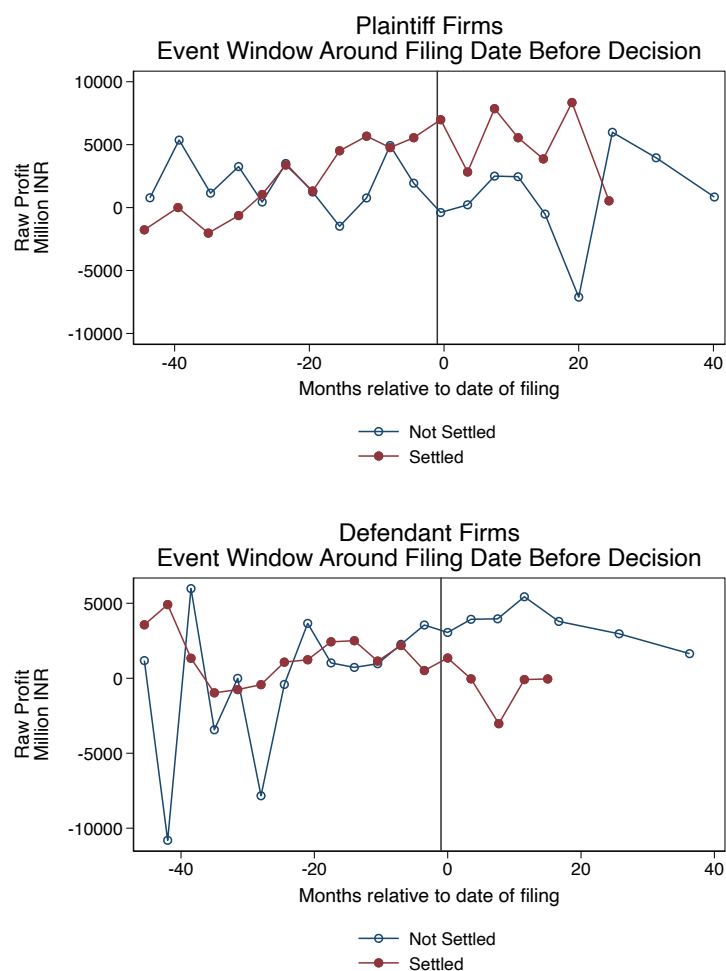


Panel B: End of Study Period



Notes: The figures present the distribution of quarterly profits across plaintiff and defendant firms in the period prior to random case assignment of judges that started in 2020 (Panel A) and at the end of the study period in 2024 when most of the cases are either settled through mediation or not (Panel B). The differences in means between settled and not settled groups of firms are not statistically or economically significant within the plaintiff or defendant samples in prior period. At endline, the differences in means between settled and not settled groups of firms among plaintiff is statistically significant with $p < 0.01$ with mean difference of Rs. 9515 Million. On the other hand, the differences in mean between settled and not settled groups among defendant firms is not statistically significant and moderately negative at endline.

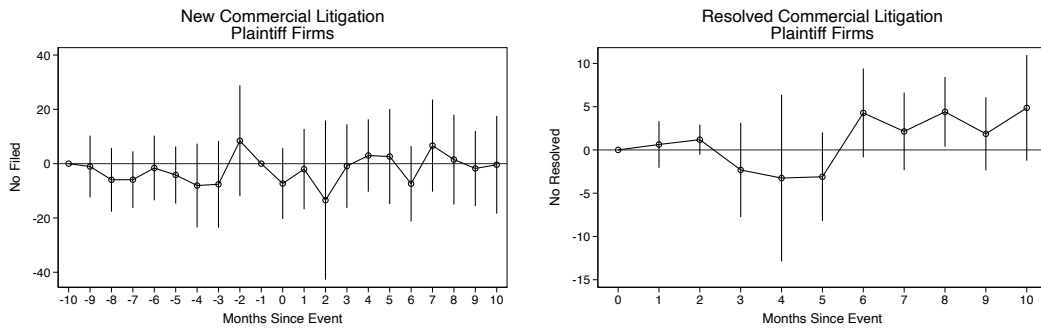
Figure A4: Raw Means of Firm-level Quarterly Profit Around Case Filing



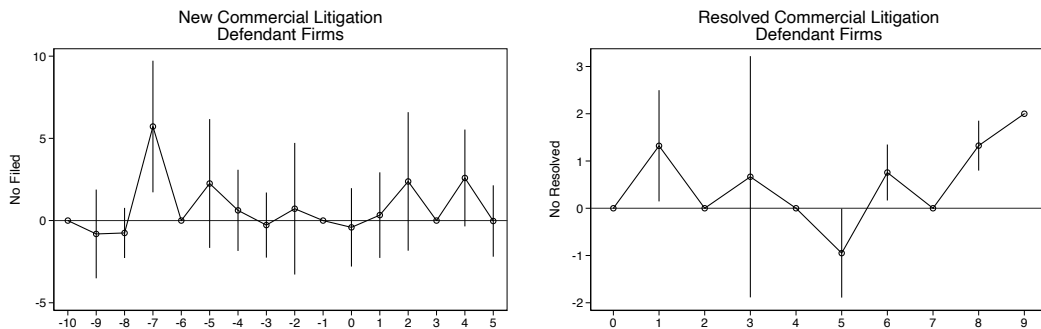
Notes: The figures above present quarterly profit, measured in standard deviation units relative to the long run average firm-specific profit, separately for plaintiff and defendant samples. The groups include those that settled during mediation and those that did not settle but obtained a case outcome during trial. Vertical lines denote the event time reference (1 quarter prior to filing). The event time is relative to the date of filing a case in the commercial court.

Figure A5: Subsequent Firm Lawsuit Behavior

Panel A: Plaintiff Firms

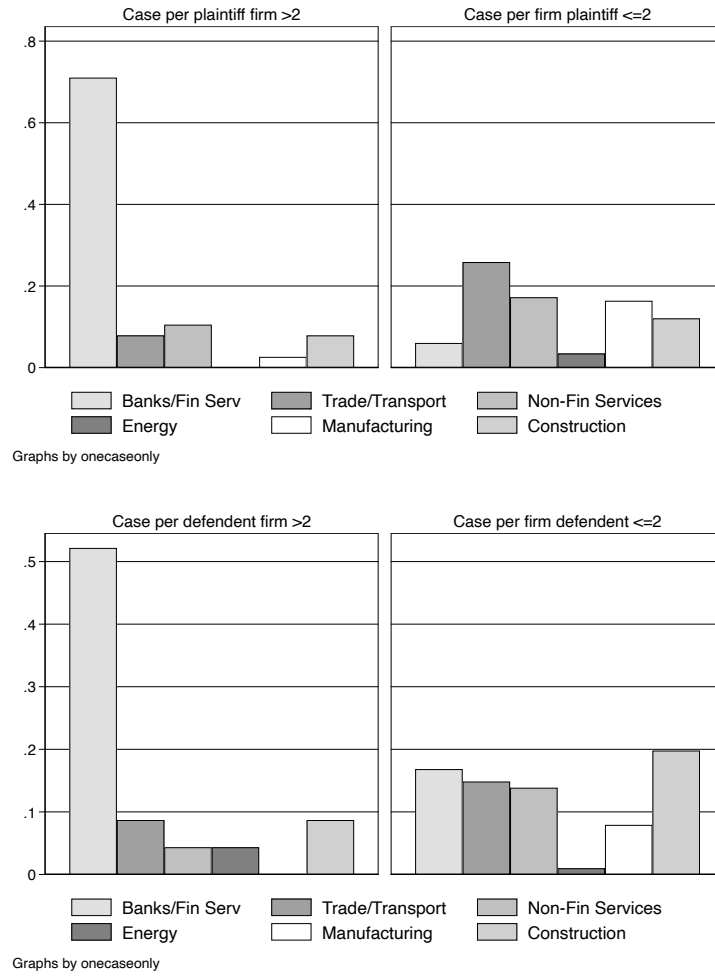


Panel B: Defendant Firms



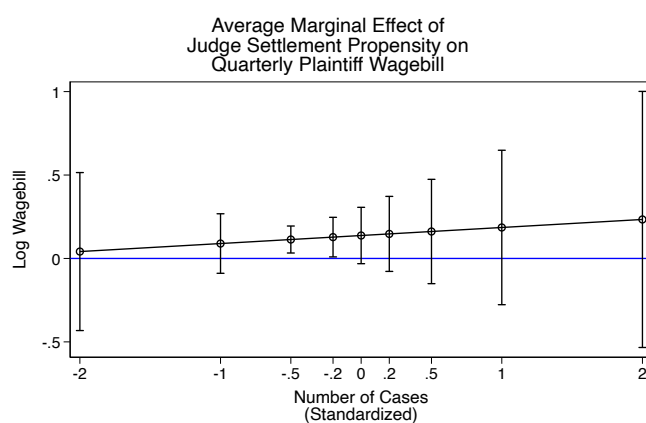
Notes: The figures above present number of cases filed and number of cases resolved by each firm in the plaintiff and defendant firm samples around the time of their first case resolution. The “treated” group are firms with cases settled through mediation. The event time is relative to the date of resolution of the first case via mediation. The control sample includes plaintiff firms with no cases settled through mediation as well as subsequent cases that were resolved through mediation later. Note that because the event is defined as first case resolution via mediation, there is no pre-period for number of subsequent case resolutions by mediation by definition.

Figure A6: Firm Sector By Number of Cases per Firm

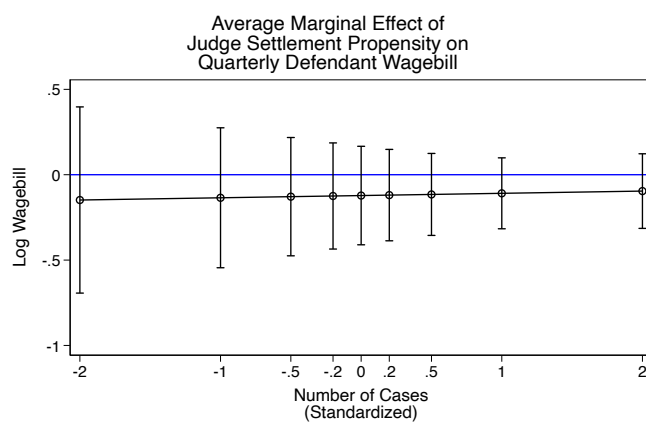


Notes: The figures above depict the distribution of firms' sector based on their litigation intensity, that is, the number of cases per firm for the plaintiff (top) and defendant (bottom) samples, respectively. For each sample, the panels present the sectoral distribution of the litigant with more than 2 cases per firm and those with 2 or fewer cases. Non-Fin services include all other service sectors other than financial services such as restaurants, salons, entertainment, etc.

Figure A7: Average Marginal Effects on Firm Wagebill
 Panel A: Plaintiff Firms



Panel B: Defendant Firms



Notes: The figures above present average marginal effects by firms' number of cases in the commercial courts, separately for plaintiff (Panel A) and defendant firms (Panel B). The dependent variable is log wage bill.

D Appendix Tables

Table A1: Case Type Summary Statistics

Distribution of Case Types			
	(1)	(2)	(3)
	Full Sample	Firm Plaintiff	Firm Defendant
Execution	0.232 (0.422)	0.208 (0.406)	0.086 (0.280)
Appeals	0.099 (0.299)	0.059 (0.235)	0.070 (0.255)
Original/New Suit	0.482 (0.500)	0.660 (0.474)	0.325 (0.469)
Other	0.187 (0.390)	0.073 (0.260)	0.520 (0.500)
Observations	16,098	3,417	502

Notes: Panel A reports percentages of litigant-type pairs in the data. Col 1 presents the distribution across the universe of commercial cases from the two courts in our study. Col 2 presents the distribution among a subset of these cases where we identify the plaintiff as a firm (either a bank/financial firm or non-financial firms). Col 3 presents the distribution among another subset where we identify the respondents as a firm (bank/financial or non-financial firms).

Table A2: Settlement propensity not correlated with quality or initial claim

	(1) Characters Per Page Order	(2) No. Precedents Order	(3) Case Value (100,000 INR)	(4) Recovery Rate
Leave-out Settled (SD)	28.29 (26.09)	0.0149 (0.0164)	-0.526 (0.328)	-0.214 (0.980)
Observations	1091	1091	464	392
No. Judges	24	24	19	15
City FE	Y	Y	Y	Y
Registration-Month FE	Y	Y	Y	Y
Case-Type FE	Y	Y	Y	Y
Control Mean	1565.4	0.613	15.98	100
Control SD	386.8	0.495	6.353	0
Adj R-Squared	0.402	0.312	0.342	0.176
F-stat	1.175	0.827	2.570	0.0476

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: This table uses case order data to estimate the correlation between the leave-out settlement propensity and case-level outcomes measuring quality of resolution (precedents) and initial claim. Standard errors are clustered by judge.

Table A3: Complier Characteristics

Panel A: Plaintiff

	Sample Mean	Complier Mean	Ratio
Few Cases (≤ 2)	.094	.108	1.154
Many Cases (> 2)	.906	.824	.908
Financial Firm	.887	.822	.927
Non-Financial Firm	.113	.117	1.032
Age ≤ 30	.151	.128	.845
Age > 30	.849	.865	1.019
Listed	.814	.796	.978
Not Listed	.186	.180	.969

Panel B: Defendant

	Sample Mean	Complier Mean	Ratio
Few Cases (≤ 2)	.467	.652	1.397
Many Cases (> 2)	.533	.544	1.020
Financial Firm	.467	.211	.451
Non-Financial Firm	.533	.540	1.013
Age ≤ 30	.518	.629	1.216
Age > 30	.482	.706	1.463
Listed	.420	.627	1.495
Not Listed	.580	.567	.976

Notes: Complier mean calculated as Sample Mean * (Subgroup First Stage / Overall First Stage). The sample contained the matched firm-case level data for plaintiff and defendant firms respectively. There are 1470 and 235 cases in the plaintiff and defendant samples, respectively. These map to 154 and 124 unique firms in each of the datasets.

Table A4: Settlement-prone judges do not decide trials differently

	(1) Plaintiff Wins	(2) Plaintiff Wins	(3) Financial Firm Wins
Leave-out Settled (SD)	0.0599*** (0.0171)	0.0525** (0.0205)	0.0349 (0.0227)
Trial=1		-0.0595 (0.0617)	-0.146*** (0.0416)
Trial=1 $\times$ Leave-out Settled (SD)		0.0115 (0.0387)	0.0175 (0.0376)
Observations	1091	1091	1091
No. Judges	24	24	24
City FE	Y	Y	Y
Registration-Month FE	Y	Y	Y
Case-Type FE	Y	Y	Y
Control Mean	0.419	0.419	0.677
Control SD	0.502	0.502	0.475
Adj R-Squared	0.529	0.530	0.491
F-stat	12.21	5.737	9.539

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: This table uses case order data to examine whether judges by their leave-out settlement propensity differentially affect plaintiff-wins or bank-wins in trials (when settlement fails). Standard errors are clustered by judge.

Table A5: Mediation is a dispute resolution strategy for banks and financial-sector firms

	(1) Goes to Trial	(2) Bank Wins	(3) Goes to Trial	(4) Bank Wins
Plaintiff Bank	-0.201*** (0.0463)	0.408*** (0.0243)		
Defendant Bank	-0.0204 (0.0851)	0.762*** (0.0717)		
Plaintiff Bank=1			-0.170*** (0.0437)	0.386*** (0.0235)
Leave-out Settled (SD)			-0.0426 (0.0326)	0.0316 (0.0224)
Plaintiff Bank=1 $\times$ Leave-out Settled (SD)			-0.0130 (0.0312)	-0.00276 (0.0416)
Defendant Bank=1			0.0196 (0.173)	0.875*** (0.107)
Defendant Bank=1 $\times$ Leave-out Settled (SD)			0.0418 (0.0831)	0.0496 (0.0520)
Observations	1097	1097	1091	1091
No. Judges	28	28	24	24
City FE	Y	Y	Y	Y
Registration-Month FE	Y	Y	Y	Y
Case-Type FE	Y	Y	Y	Y
Control Mean	0.128	0.603	0.128	0.603
Control SD	0.335	0.490	0.335	0.490
Adj R-Squared	0.183	0.565	0.178	0.565

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: This table uses case order data to examine trial behavior and trial outcomes (when settlement fails) by whether the plaintiff or defendant firm is a bank/financial sector firm and the litigant's sector interacts with judge settlement propensity. Standard errors are clustered by judge.

Table A6: Robustness: Causal effects on firms using last case settled

	(1)	(2)	(3)	(4)
	Profit (SD units)	Profit (SD units) Plaintiff Post Filing Periods Only	Profit (SD units) Defendant	Profit (SD units) Defendant Post Filing Periods Only
Post Decision=1	-0.102 (0.196)	-0.347 (0.216)	-0.0150 (0.185)	-0.286 (0.222)
Post Decision=1 $\times$ Leave-out Settled	0.422 (0.249) {p=0.084}	0.598** (0.241) {p=0.059}	-1.235** (0.471) {p=0.012}	0.420 (0.602) {p=0.503}
Observations	2910	939	2051	665
No. Judges	18	18	23	23
No. Firms	71	67	56	51
City FE	Y	Y	Y	Y
Registration-Month FE	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y
Mean Dep Var	0	0	0	0
SD Dep Var	0.988	0.988	0.987	0.987
Adj R-Squared	0.138	0.299	0.0429	0.308

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: Sample includes cases that were randomly assigned to judges in commercial courts in the two cities/metropolitan areas matched with firm-level quarterly balance sheet data. Columns 1 and 3 include firm-level data from 2011 (prior to the Commercial Courts Act, 2015), with post period denoted as quarters following the decision date of the settled case (decision date for counterfactual is the date when the case is resolved through full trial or continues to be pending outcome). In the event when firms have multiple cases, we use the last resolved case as the reference. Columns 2 and 4 subsets the data to include time periods (quarters) only after the date when a case is filed, with post period defined as previously. Standard errors are clustered by assigned judge. Bootstrapped p-values in {.}.